University of the Armed Forces

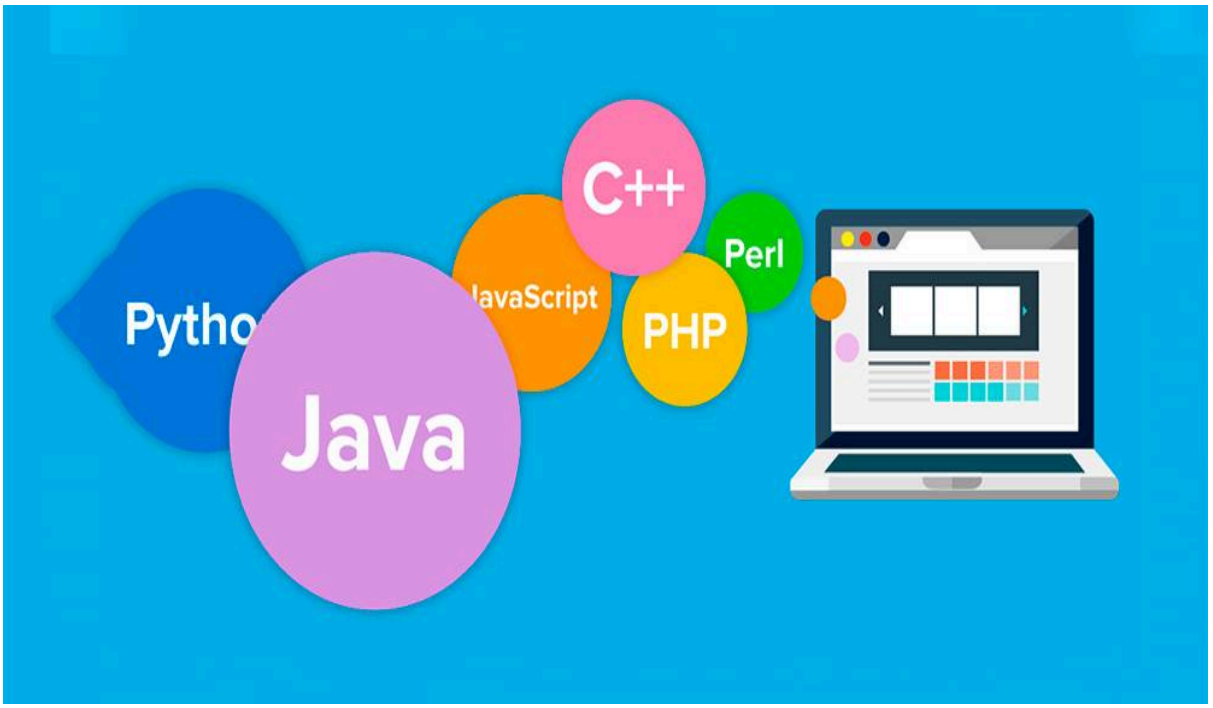
DEPARTMENT OF COMPUTER SCIENCE

Object-Oriented Programming

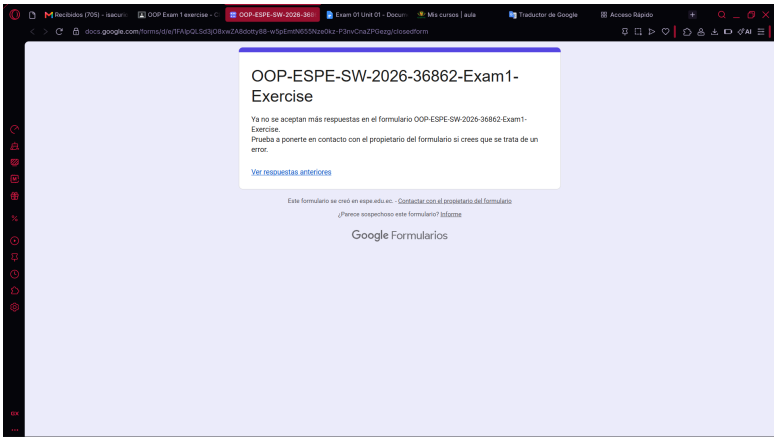
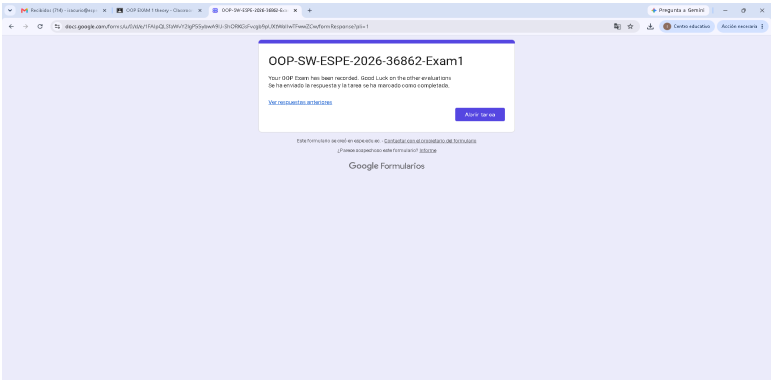
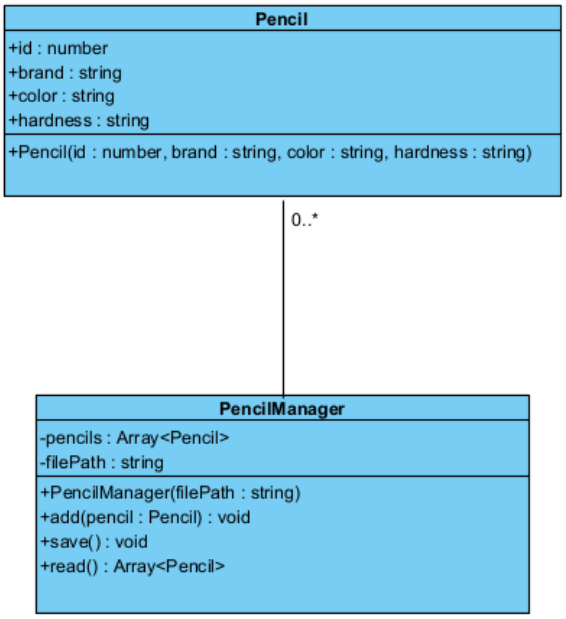
Exam 01 Unit 1

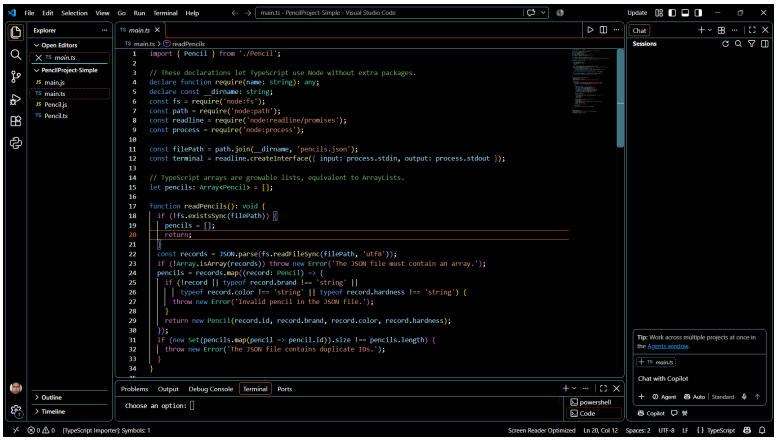
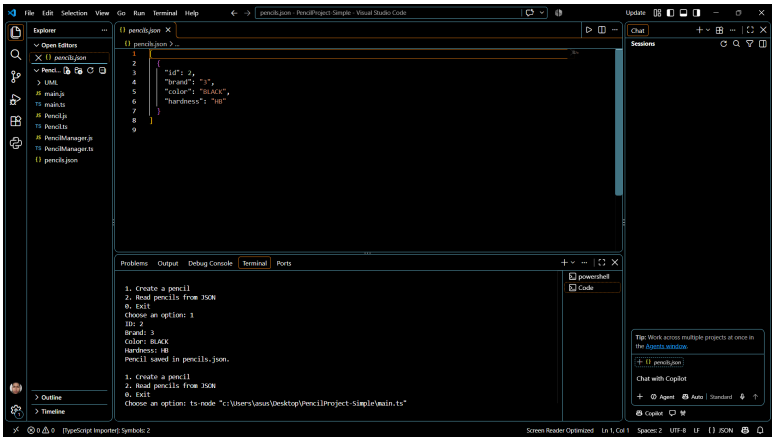
Edison Lascano PhD

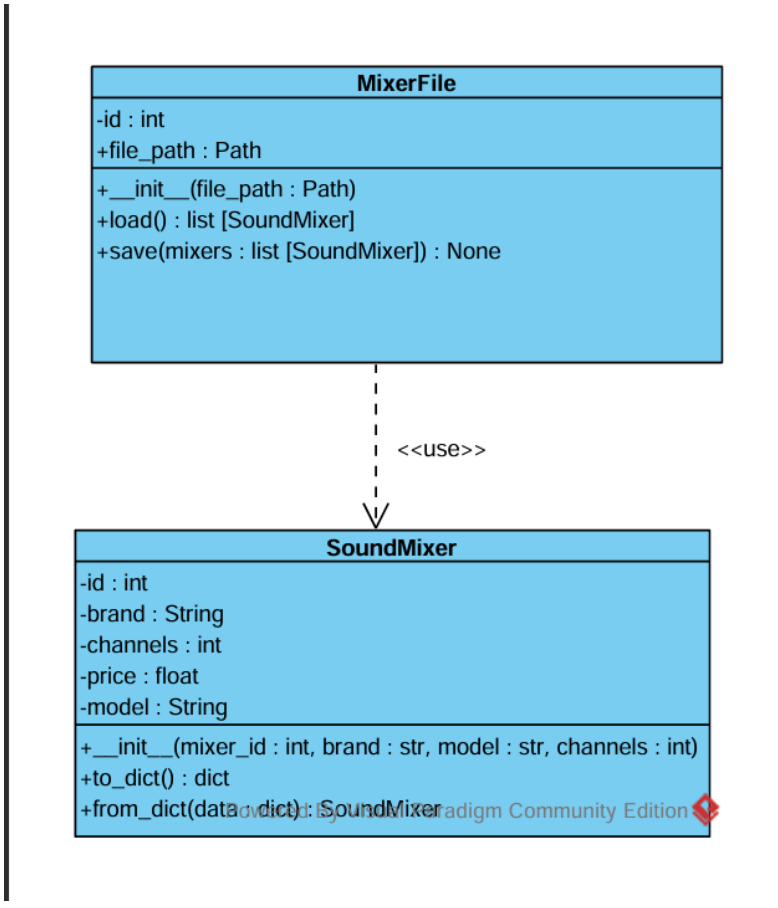
NRC: 36862

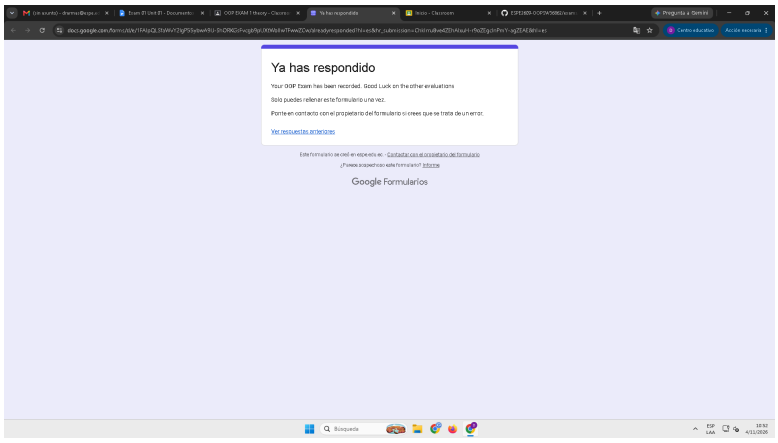


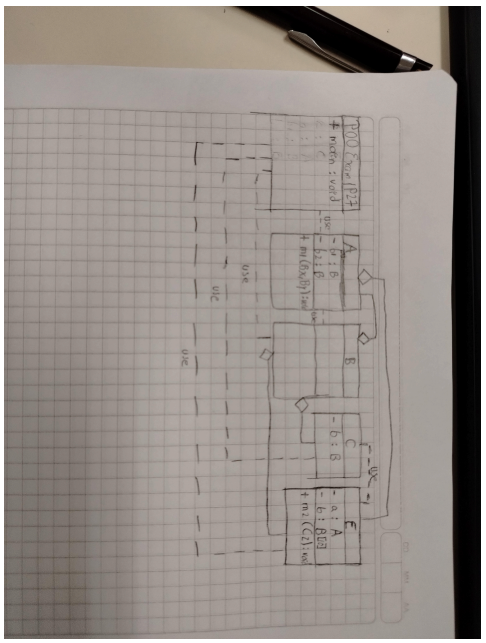
Nr.	Students		Grade

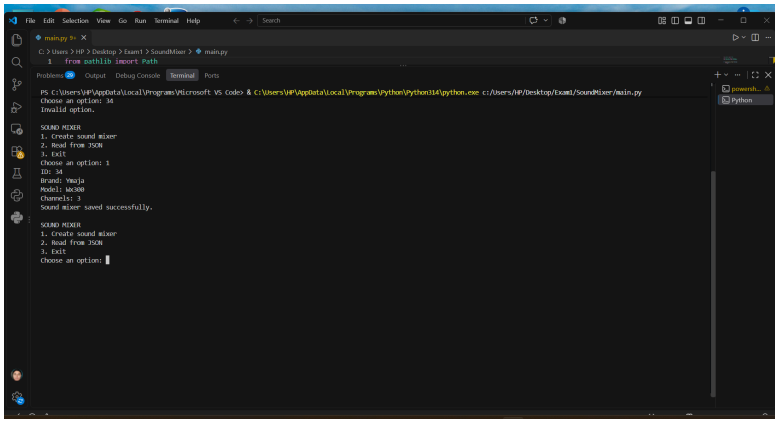
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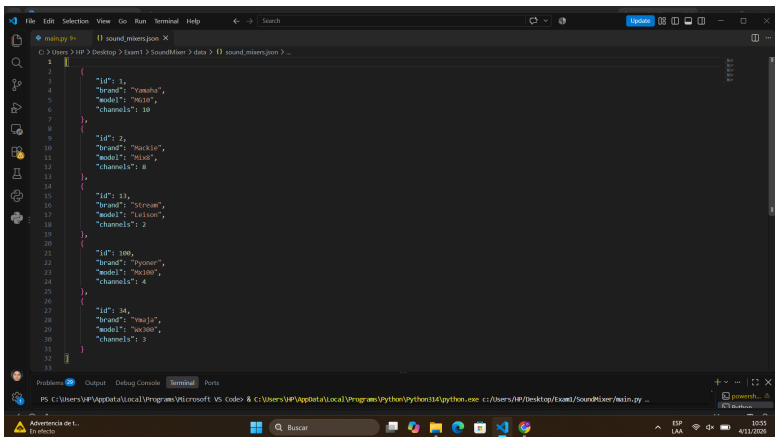
		 	
2	DEMIAN ROLAN DO ARMAS GUAÑO		

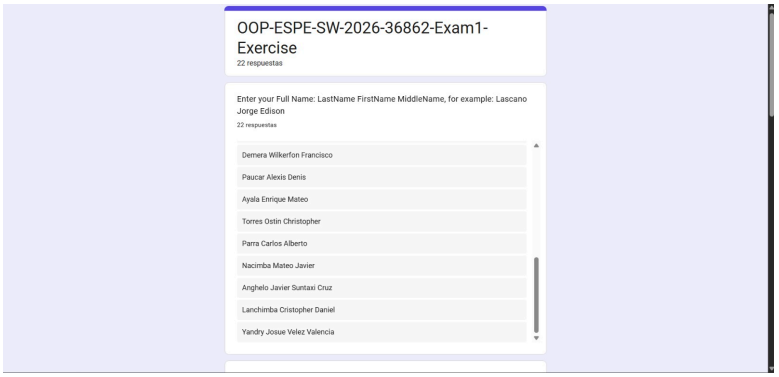
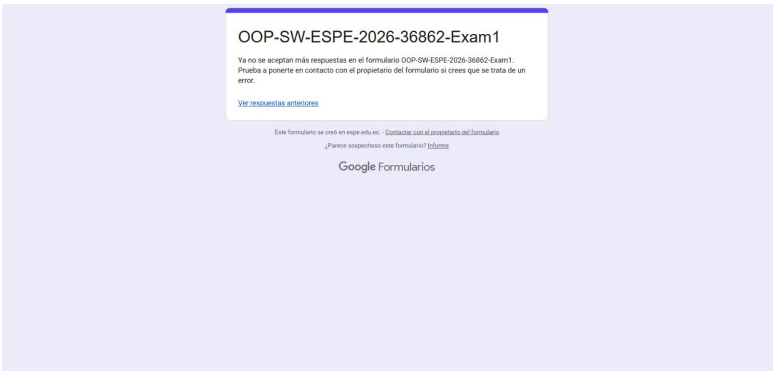
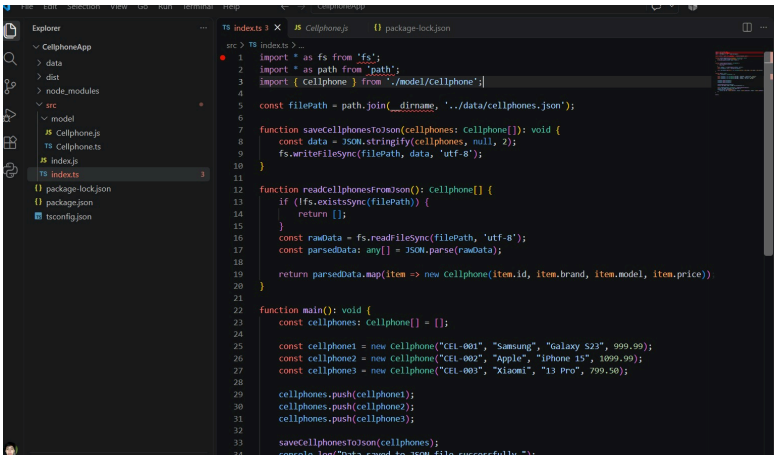

MixerFile
-id : int
+file_path : Path
+__init__(file_path : Path)
+load() : list [SoundMixer]
+save(mixers : list [SoundMixer]) : None
<<use>>
SoundMixer
-id : int
-brand : String
-channels : int
-price : float
-model : String
+__init__(mixer_id : int, brand : str, model : str, channels : int)
+to_dict() : dict
+from_dict(data : dict) : SoundMixer
Activated by PyCharm Professional Edition

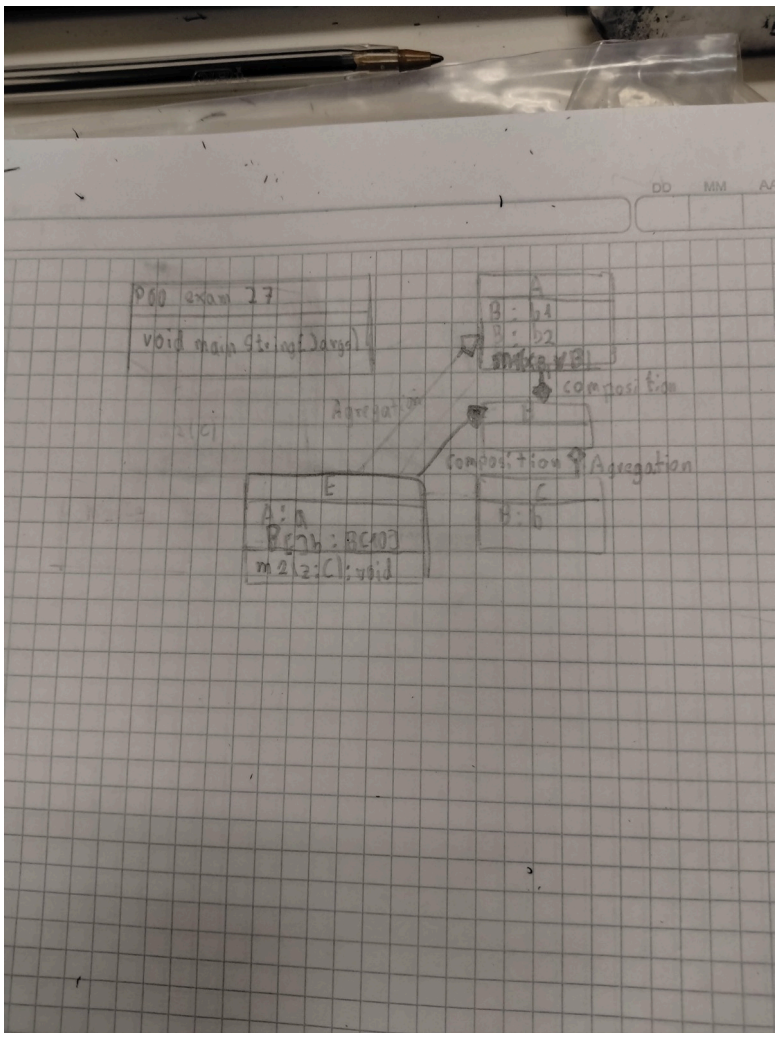

Ya has respondido
Your OOP Exam has been recorded. Good Luck on the other evaluations
Solo puedes rellenar este formulario una vez.
Ponte en contacto con el propietario del formulario si crees que se trata de un error.
Ver respuestas anteriores
Google Formularios

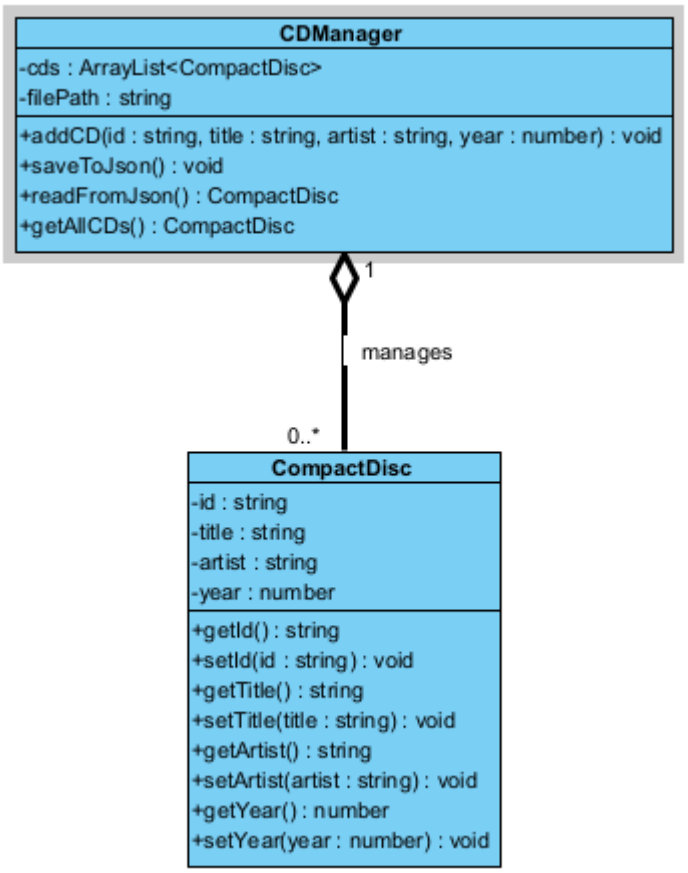



SOUND MIXER
1. Create sound mixer
2. Read from JSON
3. Exit
Choose an option: 1
ID: 34
Brand: Vmaja
Model: Wx300
Channels: 3
Sound mixer saved successfully.
SOUND MIXER
1. Create sound mixer
2. Read from JSON
3. Exit
Choose an option:


"id": 1,
"brand": "Yamaha",
"model": "MG10",
"channels": 10
"id": 2,
"brand": "Mackie",
"model": "Mix8",
"channels": 8
"id": 13,
"brand": "Stream",
"model": "Leison",
"channels": 2
"id": 100,
"brand": "Pyoner",
"model": "Mx100",
"channels": 4
"id": 34,
"brand": "Vmaja",
"model": "Wx300",
"channels": 3

3	ENRIQUE MATEO AYALA LAGUNAS	  	

			
4	ANGEL ALEJANDRO BONILLA ROMERO		
5	HENRY PATRICIO CHALCUALAN SILVA		




```typescript
import { CompactDisc } from './CompactDisc';

export class CDManager {
    private cds: CompactDisc[];
    private filePath: string;

    constructor(filePath: string) {
        this.cds = [];
        this.filePath = filePath;
    }

    public addCD(id: string, title: string, artist: string, year: number): void {
        const newCD = new CompactDisc(id, title, artist, year);
        this.cds.push(newCD);
    }

    public saveToJson(): void {
    }

    public readFromJson(): CompactDisc {
        throw new Error("Method not implemented.");
    }

    public getAllCDs(): CompactDisc {
        throw new Error("Method not implemented.");
    }
}
```

```typescript
export class CompactDisc {
    private id: string;
    private title: string;
    private artist: string;
    private year: number;

    constructor(id: string, title: string, artist: string, year: number) {
        this.id = id;
        this.title = title;
        this.artist = artist;
        this.year = year;
    }

    public getId(): string {
        return this.id;
    }

    public setId(id: string): void {
        this.id = id;
    }

    public getTitle(): string {
        return this.title;
    }

    public setTitle(title: string): void {
        this.title = title;
    }

    public getArtist(): string {
        return this.artist;
    }
```

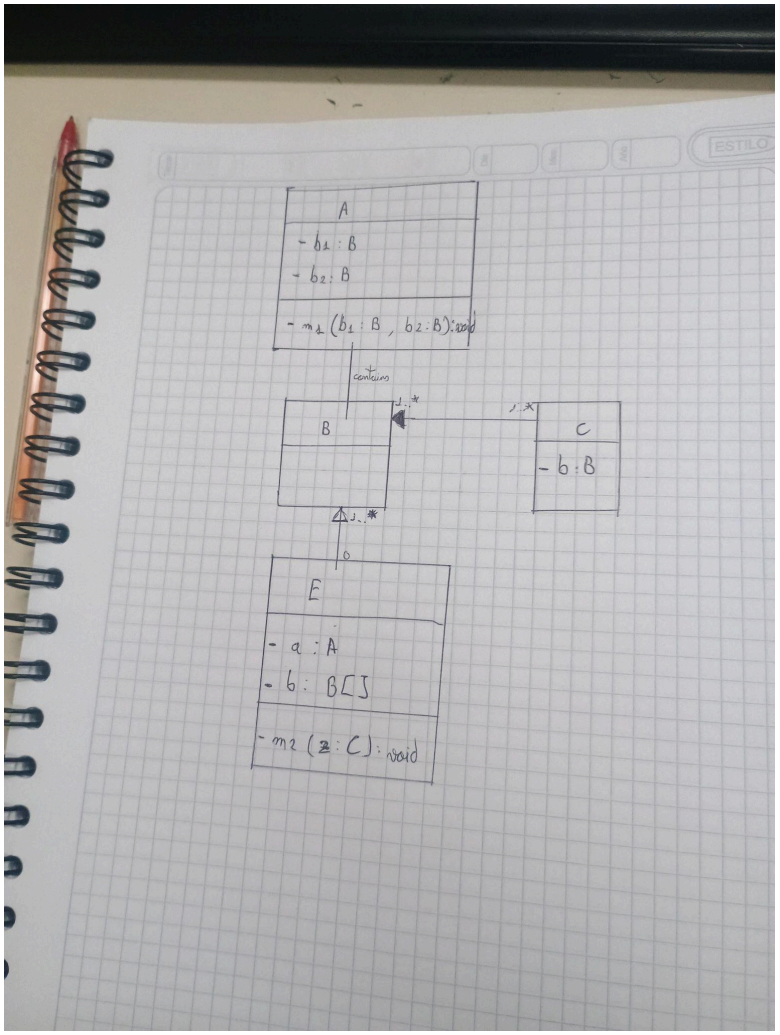
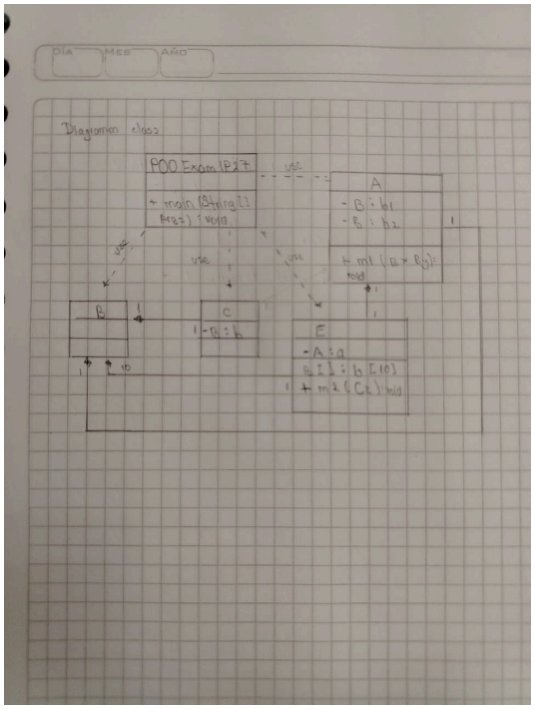
```json
{
  "name": "u1-cdmanager",
  "version": "1.0.0",
  "description": "Proyecto de gestión de CDs en TypeScript",
  "main": "main.ts",
  Debug
  "scripts": {
    "start": "ts-node main.ts",
    "build": "tsc"
  },
  "devDependencies": {
    "ts-node": "^10.9.2",
    "typescript": "^5.0.0"
  }
}
```

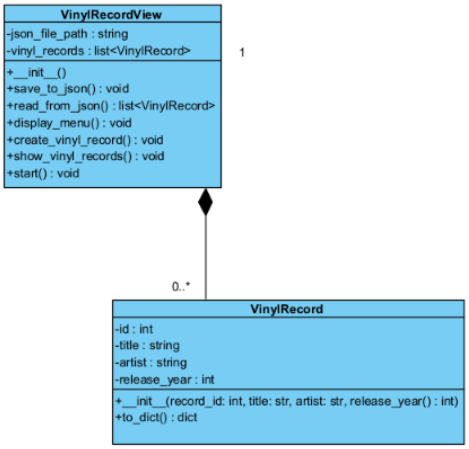
```typescript
import { CDManager } from './CDManager';

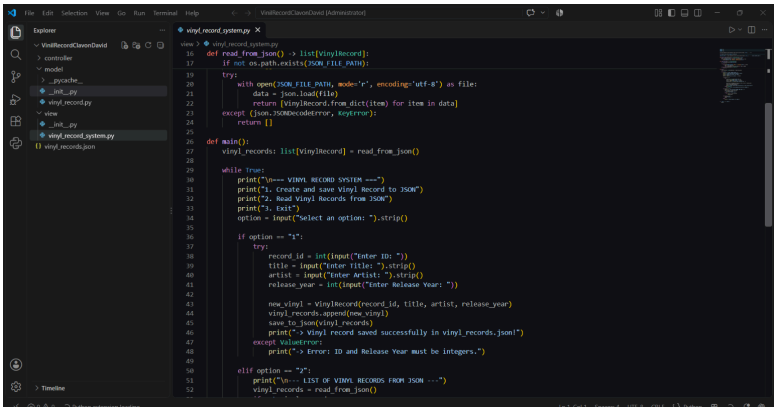
const manager = new CDManager('cds.json');

manager.addCD('1', 'The Dark Side of the Moon', 'Pink Floyd', 1973);
manager.addCD('2', 'Thriller', 'Michael Jackson', 1982);

console.log("¡CDManager inicializado y discos agregados correctamente!");
```

			
6	DAVID ALEXANDER CLAVON LOZA		





```
def read_from_json() -> list[VinylRecord]:
    if not os.path.exists(JSON_FILE_PATH):
        return []
    try:
        with open(JSON_FILE_PATH, mode='r', encoding='utf-8') as file:
            data = json.load(file)
            return [VinylRecord.from_dict(item) for item in data]
    except (json.JSONDecodeError, KeyError):
        return []

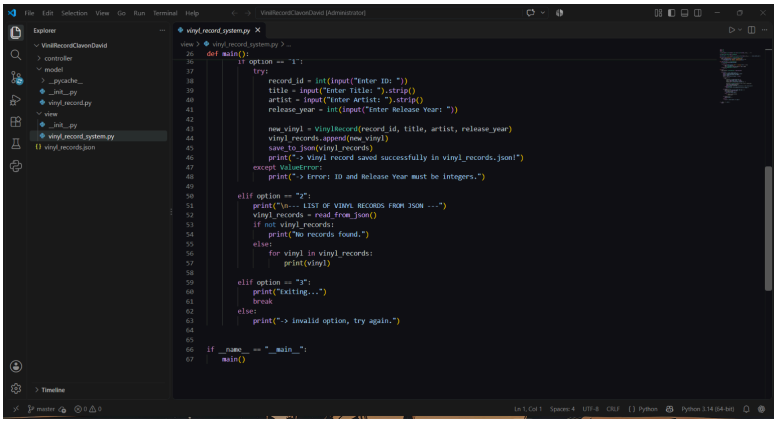
def main():
    vinyl_records: list[VinylRecord] = read_from_json()

    while True:
        print("\n=== VINYL RECORD SYSTEM ===")
        print("1. Create and save Vinyl Record to JSON")
        print("2. Read Vinyl Records from JSON")
        print("3. Exit")
        option = input("Select an option: ").strip()

        if option == "1":
            try:
                record_id = int(input("Enter ID: "))
                title = input("Enter Title: ").strip()
                artist = input("Enter Artist: ").strip()
                release_year = int(input("Enter Release Year: "))

                new_vinyl = VinylRecord(record_id, title, artist, release_year)
                vinyl_records.append(new_vinyl)
                save_to_json(vinyl_records)
                print("-> Vinyl record saved successfully in vinyl_records.json!")
            except ValueError:
                print("-> Error: ID and Release Year must be integers.")

        elif option == "2":
            print("\n--- LIST OF VINYL RECORDS FROM JSON ---")
            vinyl_records = read_from_json()
```



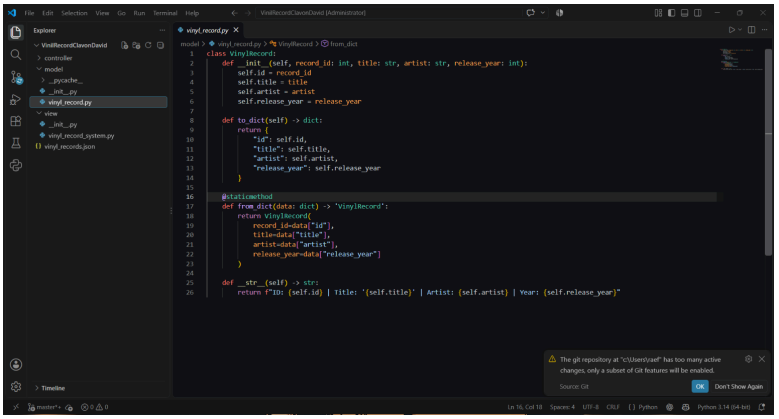
```
def main():
    if option == "1":
        try:
            record_id = int(input("Enter ID: "))
            title = input("Enter Title: ").strip()
            artist = input("Enter Artist: ").strip()
            release_year = int(input("Enter Release Year: "))

            new_vinyl = VinylRecord(record_id, title, artist, release_year)
            vinyl_records.append(new_vinyl)
            save_to_json(vinyl_records)
            print("-> Vinyl record saved successfully in vinyl_records.json!")
        except ValueError:
            print("-> Error: ID and Release Year must be integers.")

    elif option == "2":
        print("\n--- LIST OF VINYL RECORDS FROM JSON ---")
        vinyl_records = read_from_json()
        if not vinyl_records:
            print("No records found.")
        else:
            for vinyl in vinyl_records:
                print(vinyl)

    elif option == "3":
        print("Exiting...")
        break
    else:
        print("-> invalid option, try again.")


if __name__ == "__main__":
    main()
```

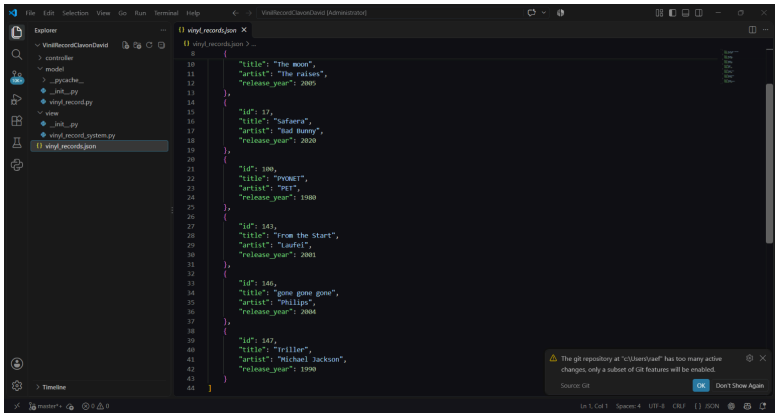
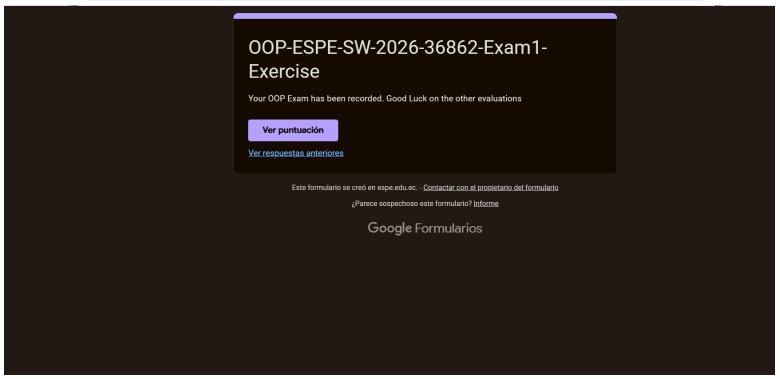
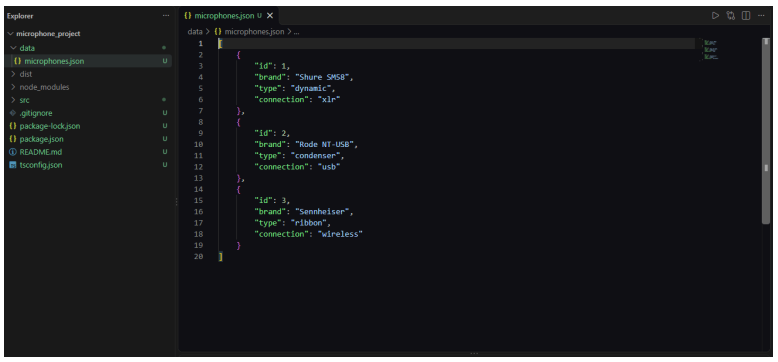


```
class VinylRecord:
    def __init__(self, record_id: int, title: str, artist: str, release_year: int):
        self.id = record_id
        self.title = title
        self.artist = artist
        self.release_year = release_year

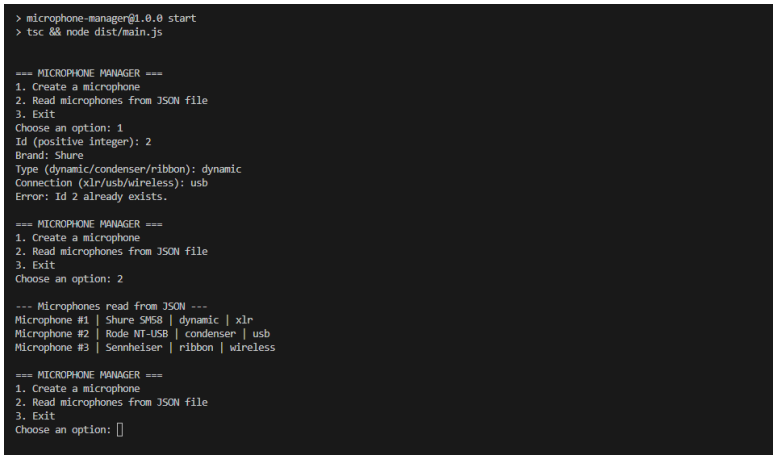
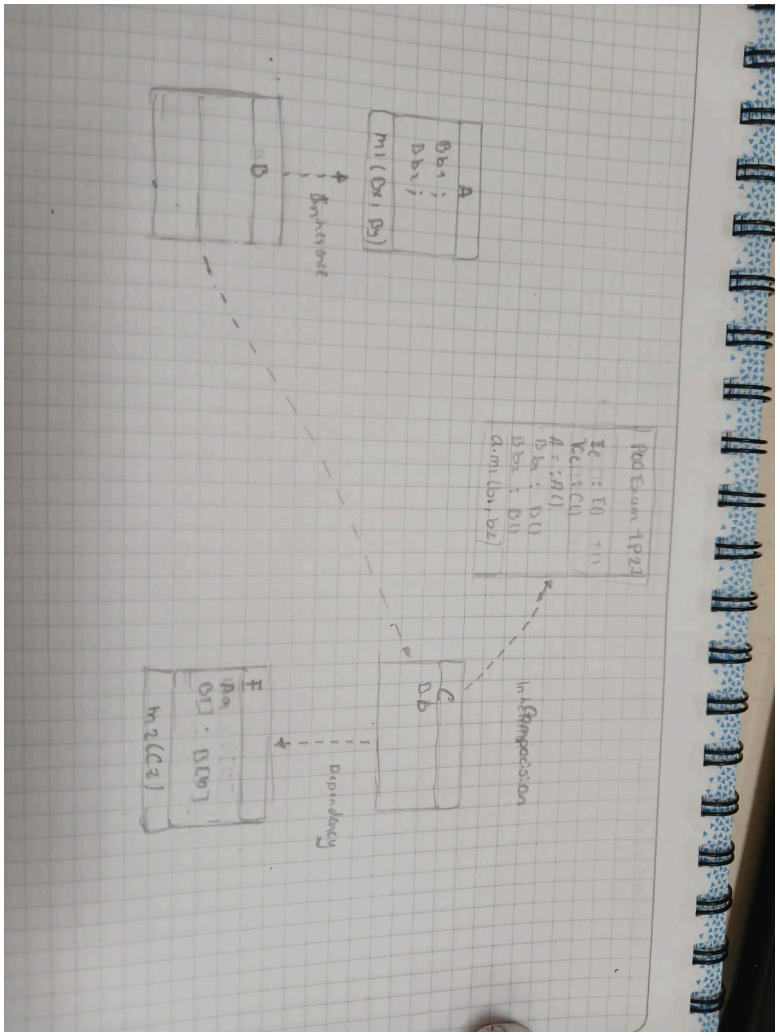
    def to_dict(self) -> dict:
        return {
            "id": self.id,
            "title": self.title,
            "artist": self.artist,
            "release_year": self.release_year
        }

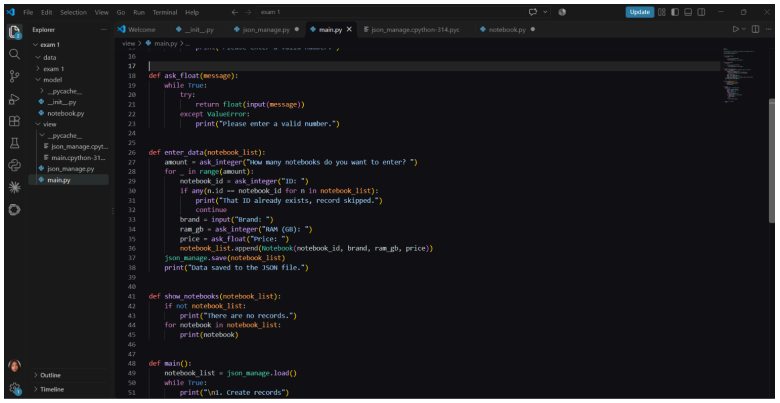
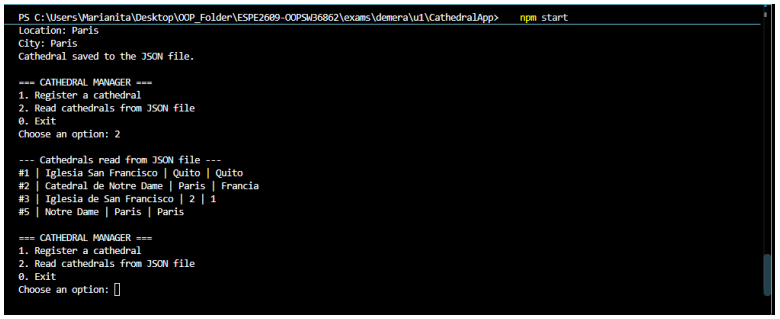
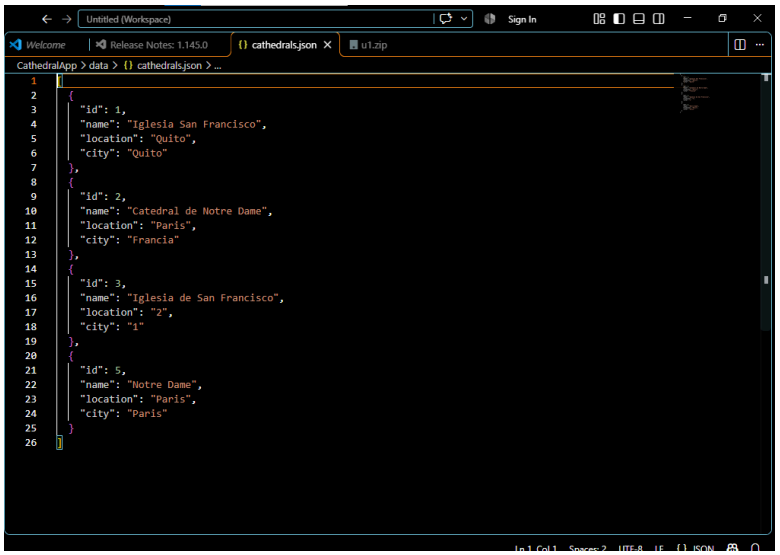
    @staticmethod
    def from_dict(data: dict) -> 'VinylRecord':
        return VinylRecord(
            record_id=data["id"],
            title=data["title"],
            artist=data["artist"],
            release_year=data["release_year"]
        )

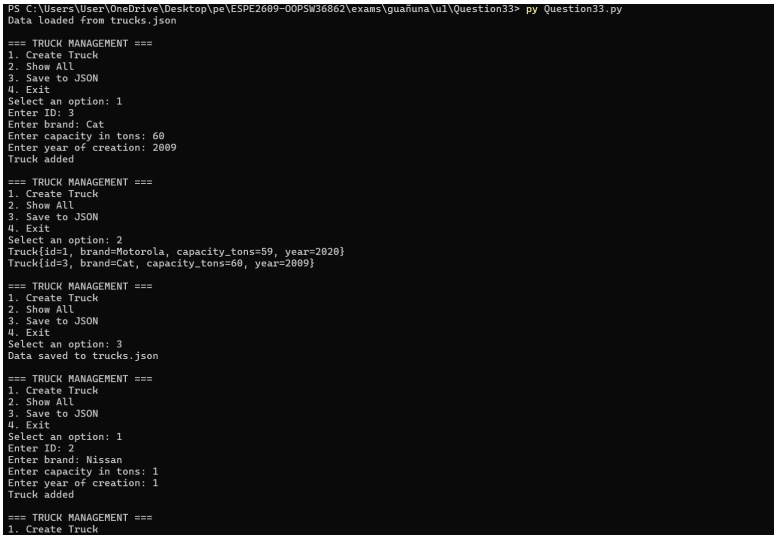
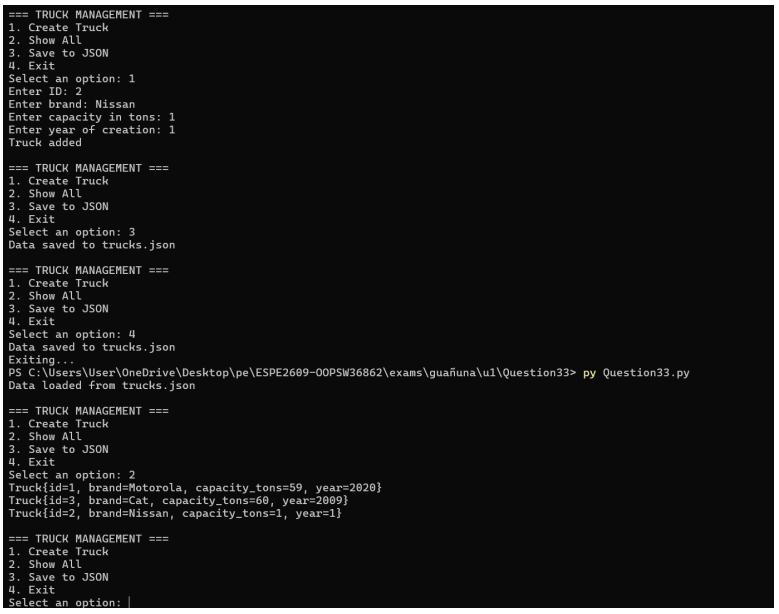
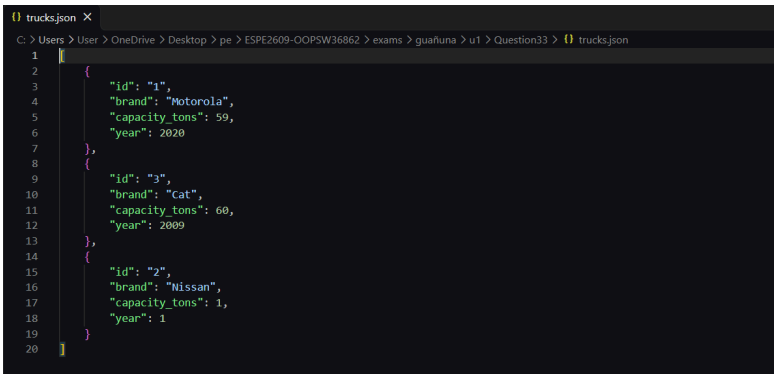
    def __str__(self) -> str:
        return f"ID: {self.id} | Title: '{self.title}' | Artist: {self.artist} | Year: {self.release_year}"
```

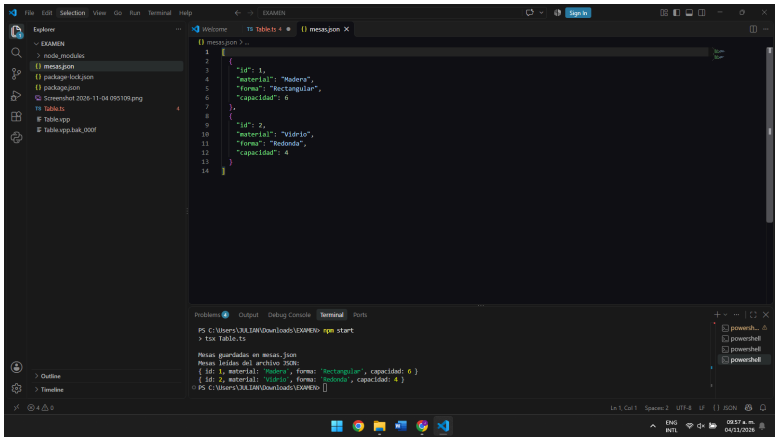
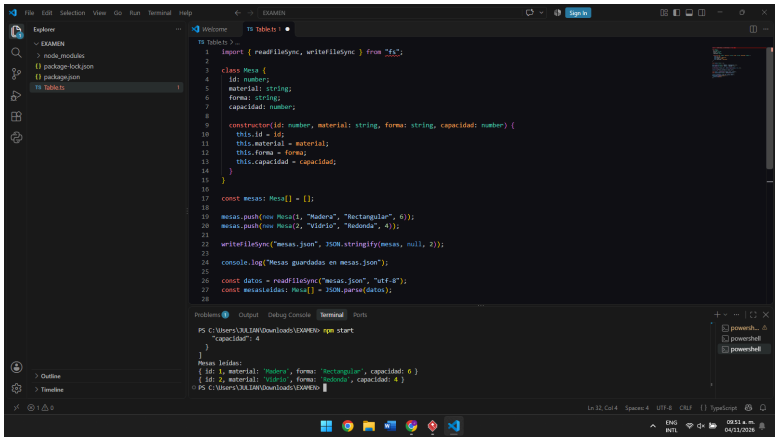
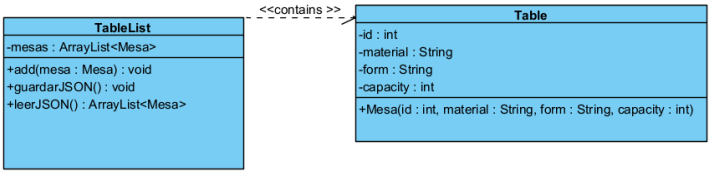
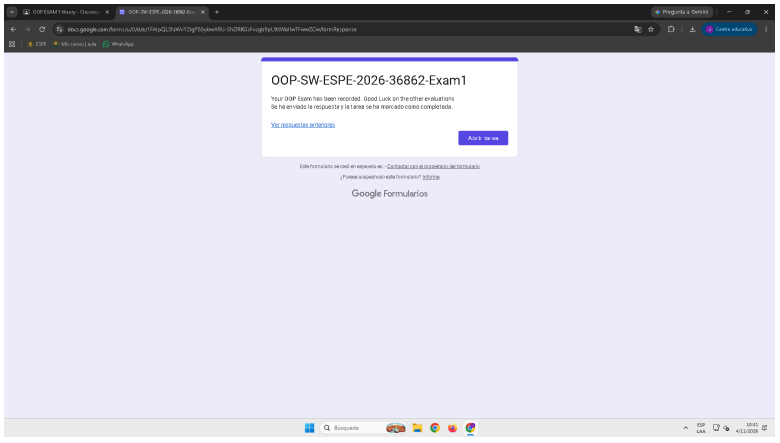
			
7	DENNIS PATRICIO CRIOLLO CHUSHIG	 	

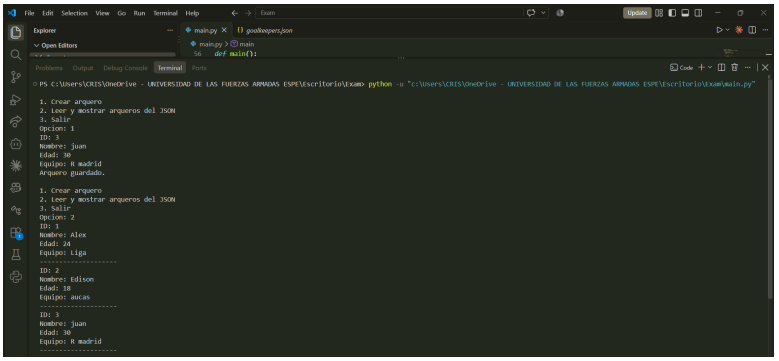
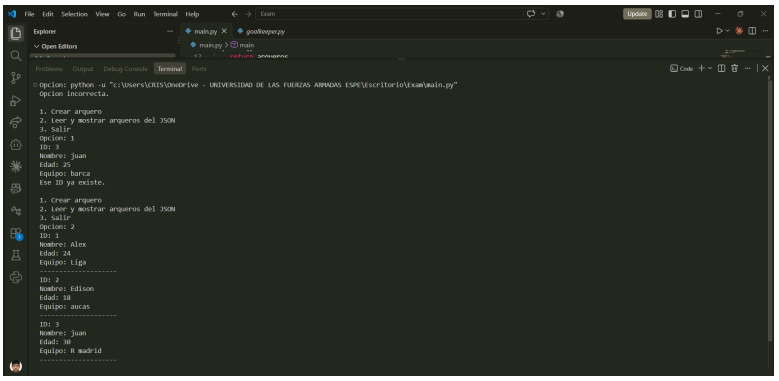
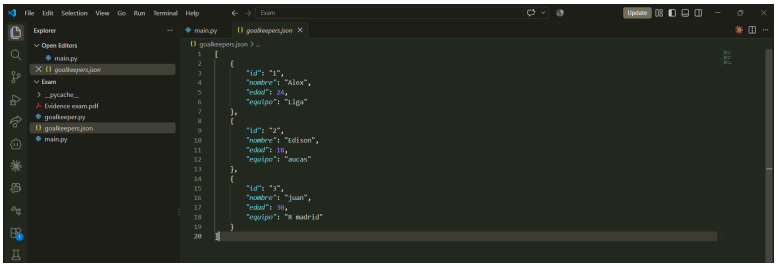
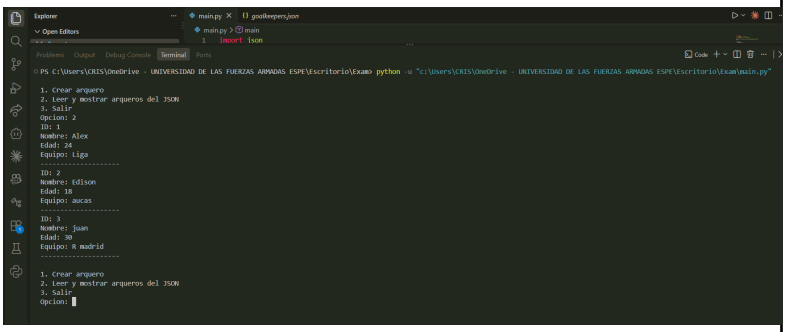
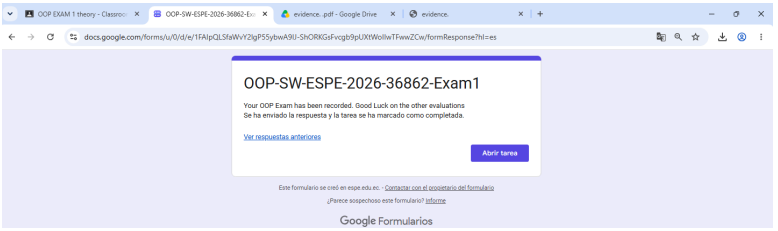

```
[
    {
        "id": 1,
        "brand": "Shure SM58",
        "type": "dynamic",
        "connection": "xlr"
    },
    {
        "id": 2,
        "brand": "Rode NT-USB",
        "type": "condenser",
        "connection": "usb"
    },
    {
        "id": 3,
        "brand": "Sennheiser",
        "type": "ribbon",
        "connection": "wireless"
    }
]
```

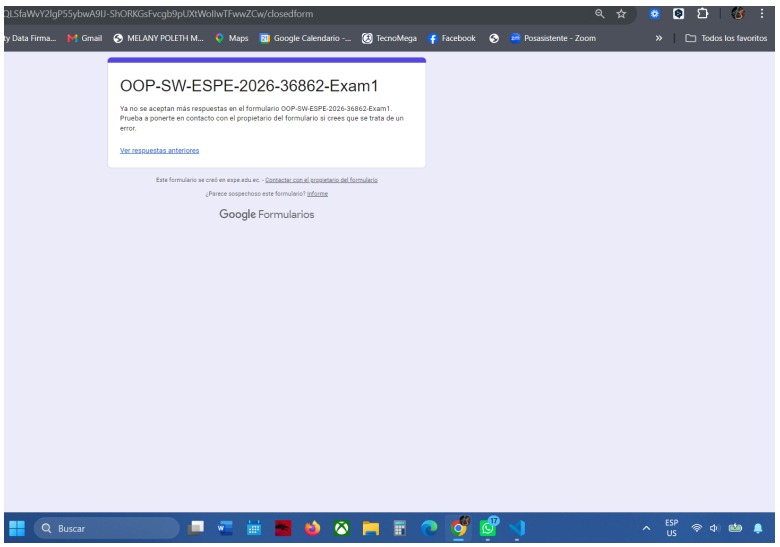
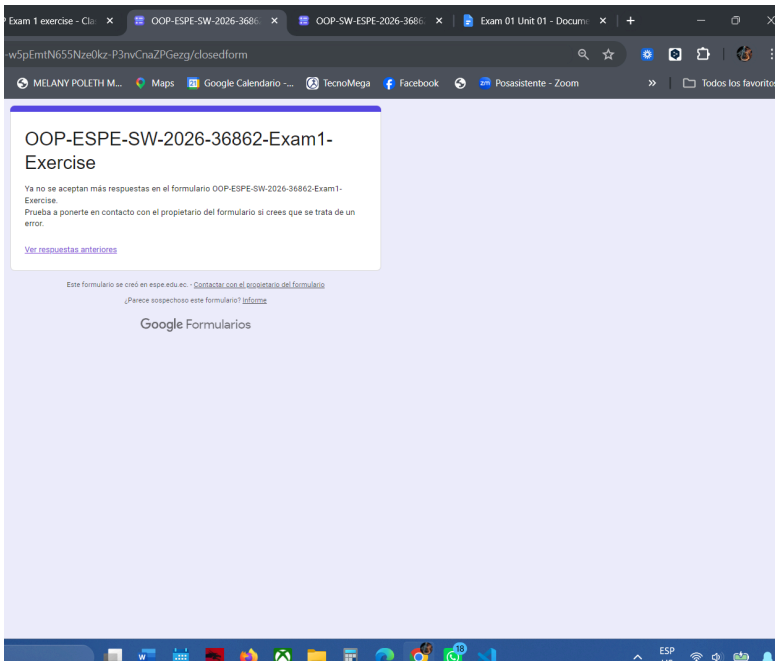
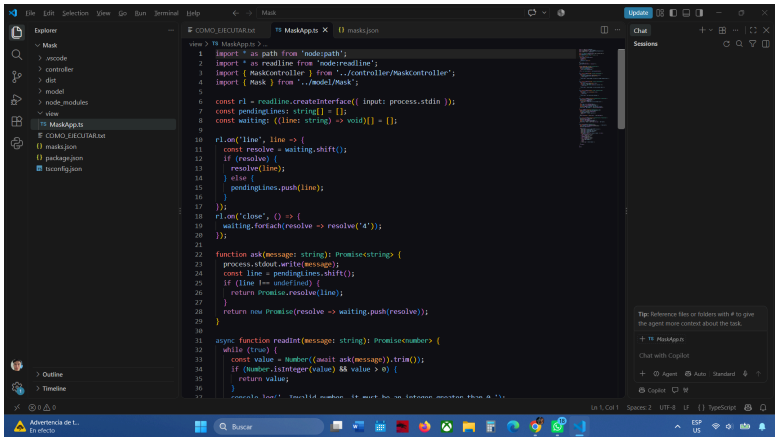
			
8	LESLYE CAMILA CUICHAN BAUTISTA		

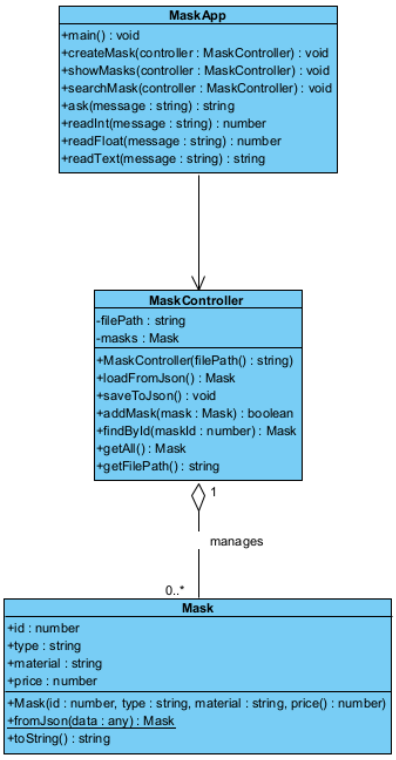
			
9	WILKE RFON FRANCI SCO DEMER A REYNA	 	

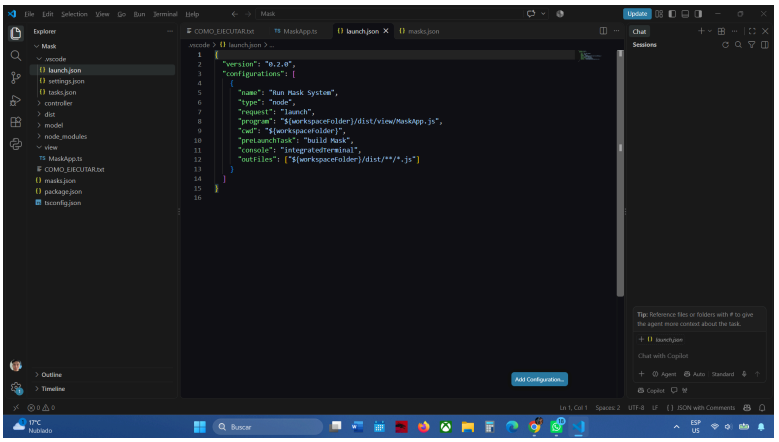
10	CRISTOPHER JOHN GUAÑUNA AJILA	  	+

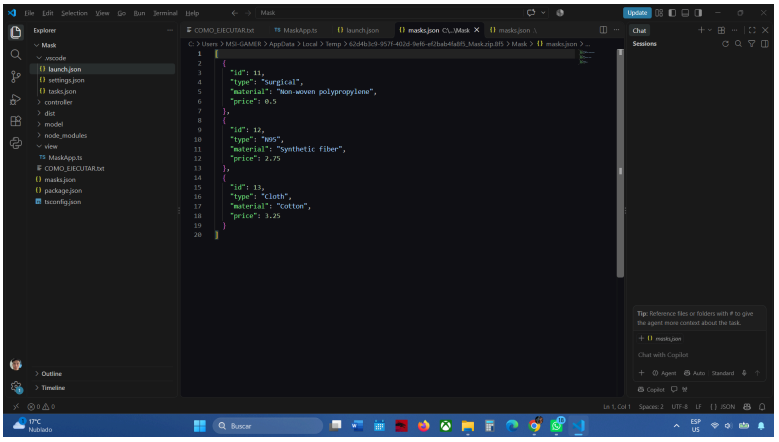
11	JULIAN ALFONSO JARAMILLO NAUPARI	    	

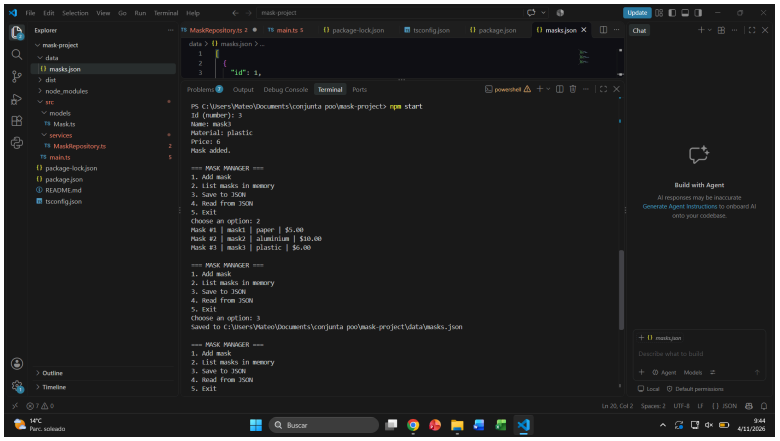
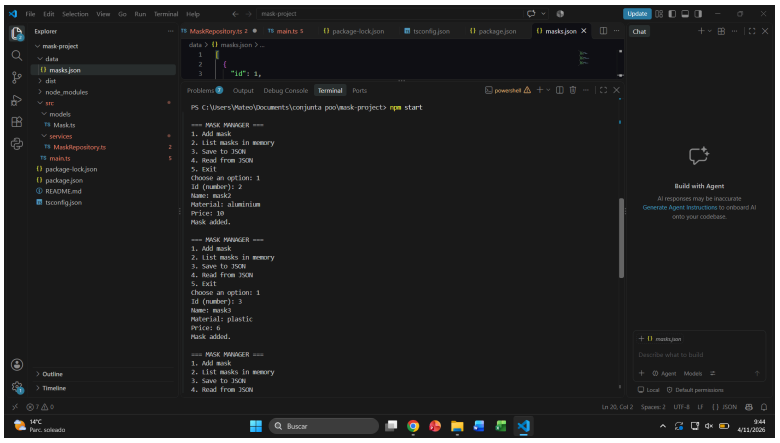
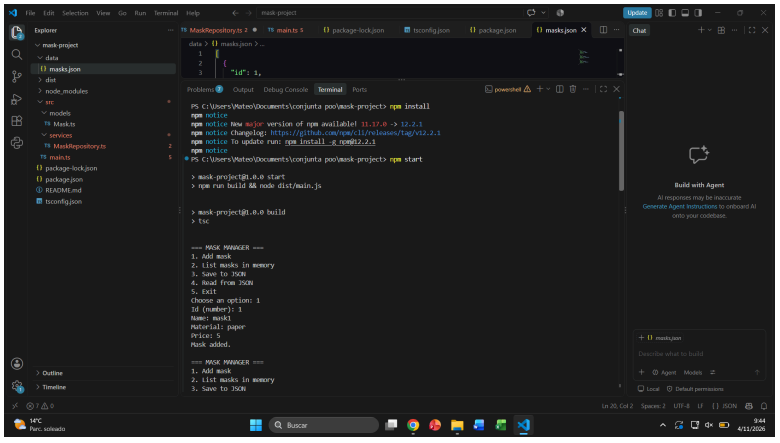
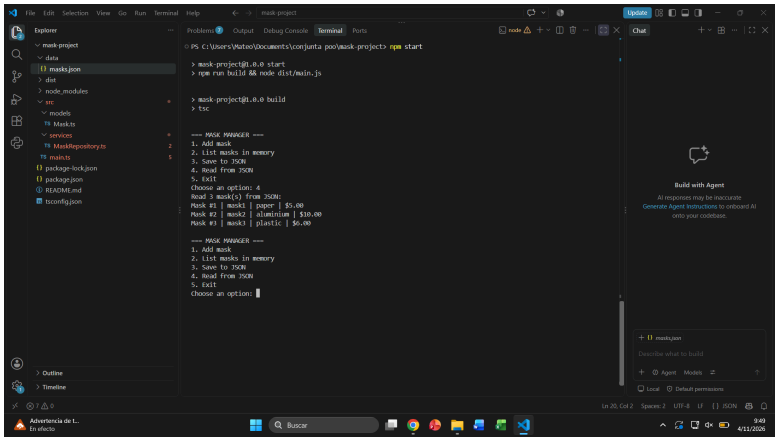
12	CRISTOPHER DANIEL LANCHIMBA CARVAJAL	    	
13	MARCOS ARIEL MAMA		

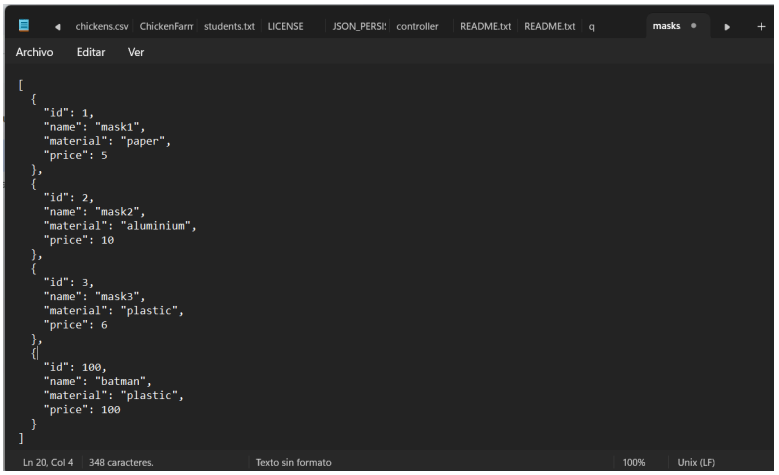
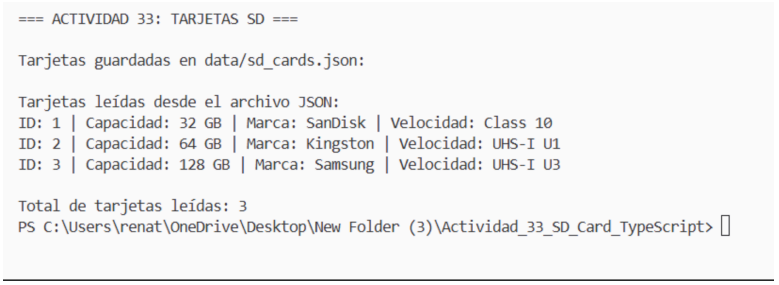
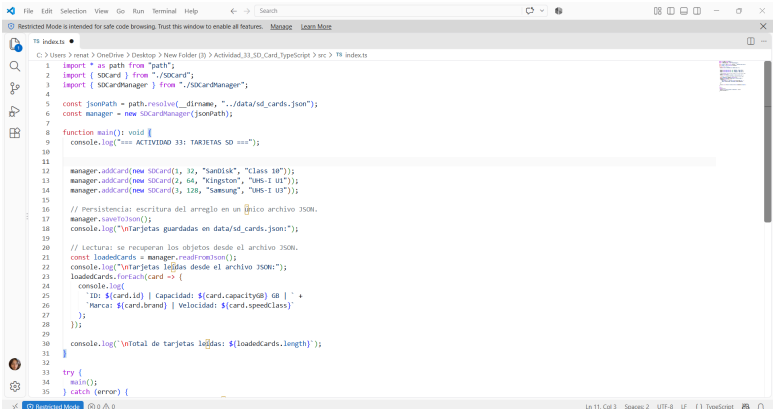
	RANDI TAPIA	  	

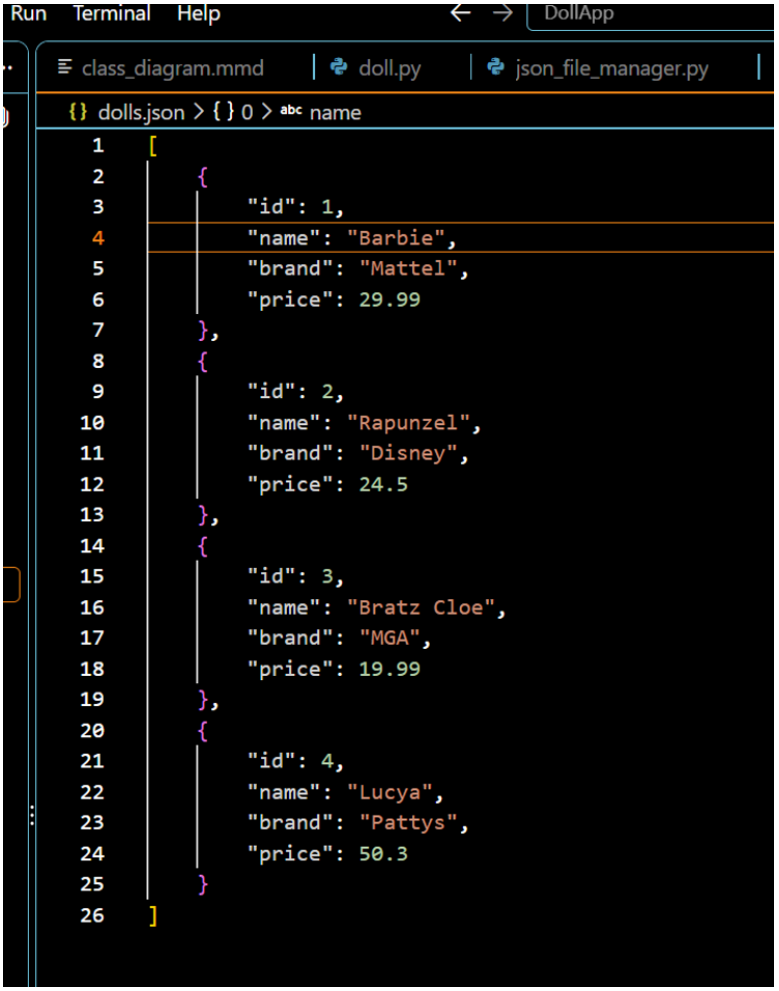

MaskApp
+main() : void
+createMask(controller : MaskController) : void
+showMasks(controller : MaskController) : void
+searchMask(controller : MaskController) : void
+ask(message : string) : string
+readInt(message : string) : number
+readFloat(message : string) : number
+readText(message : string) : string
MaskController
-filePath : string
-masks : Mask
+MaskController(filePath() : string)
+loadFromJson() : Mask
+saveToJson() : void
+addMask(mask : Mask) : boolean
+findById(maskId : number) : Mask
+getAll() : Mask
+getFilePath() : string
1
manages
0..*
Mask
+id : number
+type : string
+material : string
+price : number
+Mask(id : number, type : string, material : string, price() : number)
+fromJson(data : any) : Mask
+toString() : string

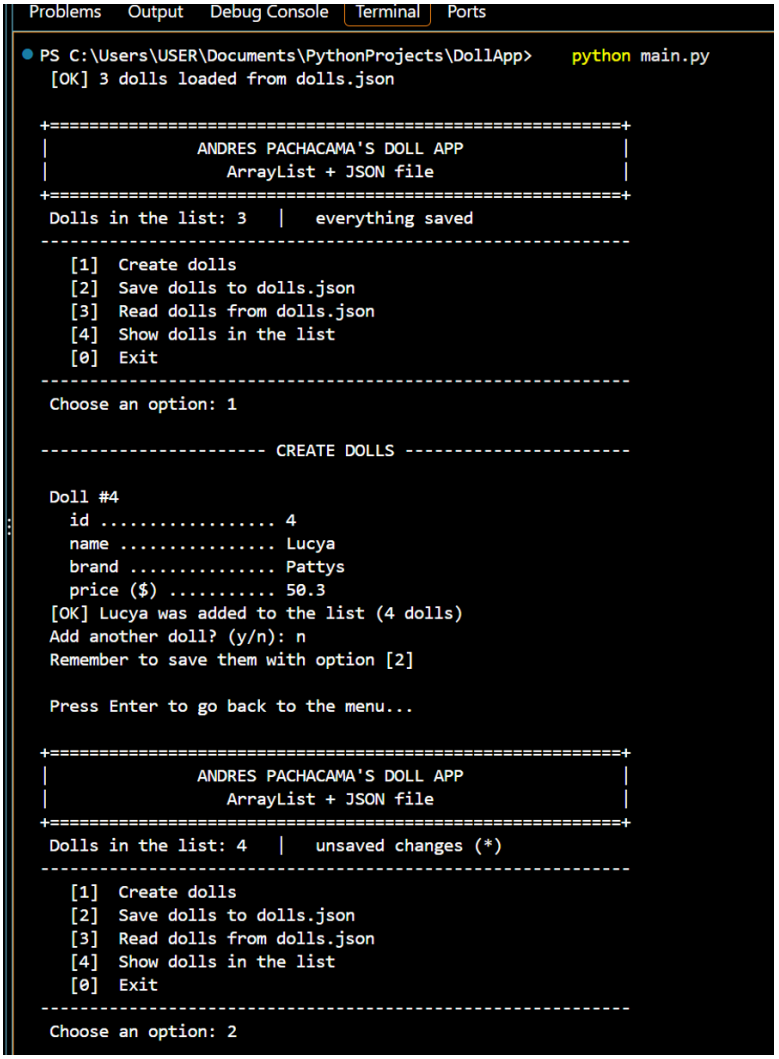


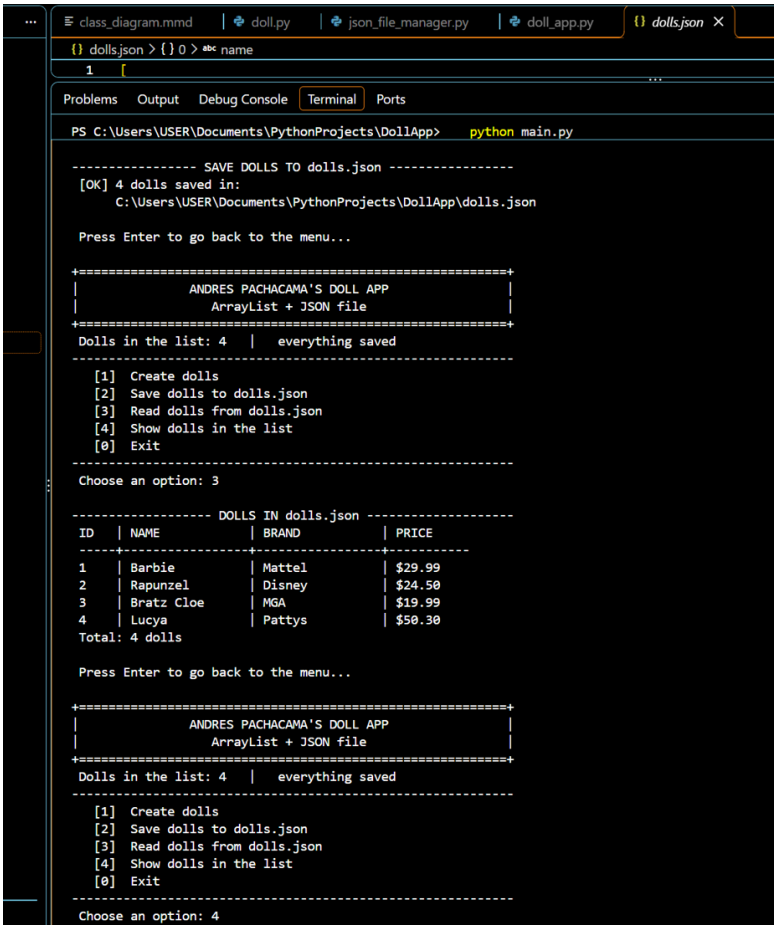


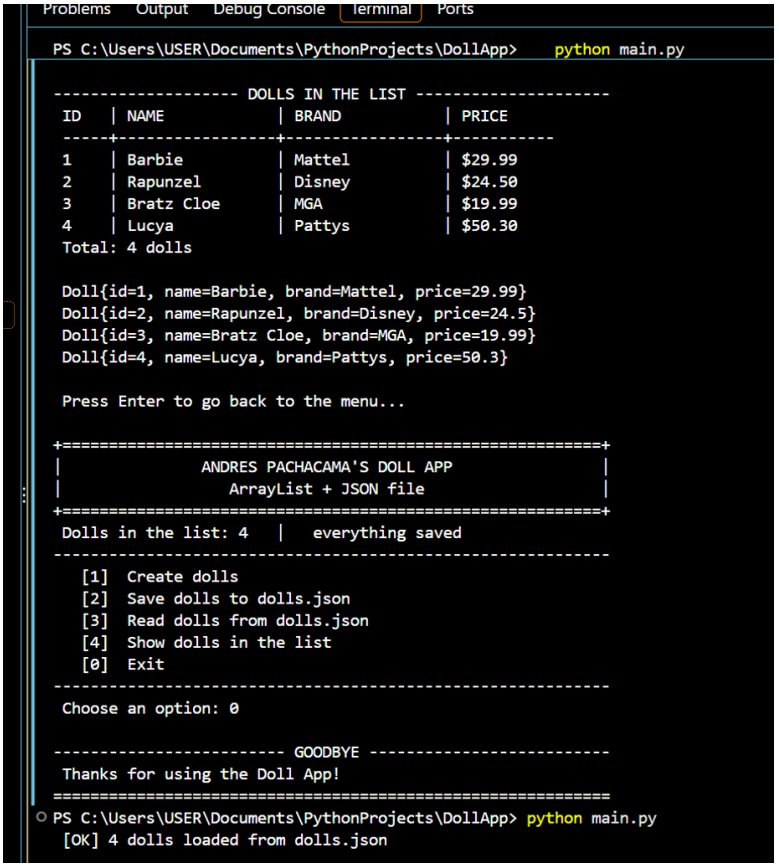
14	MATEO JAVIER NACIM BA ÑATO	   

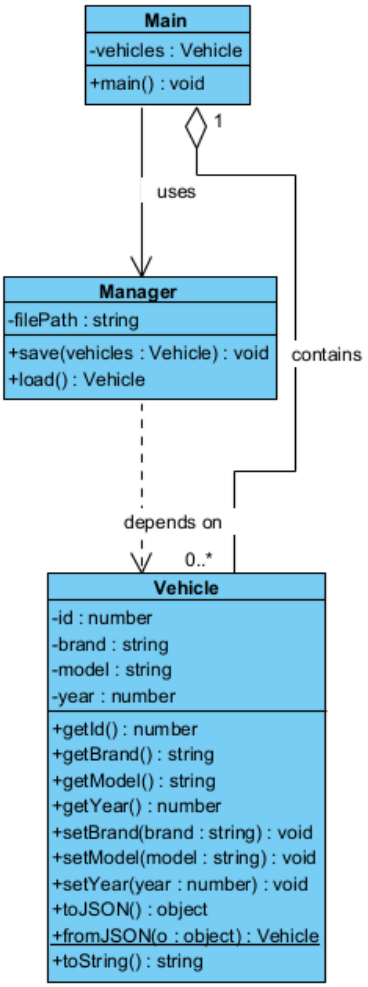
			
15	RENATA ABIGAIL OLALLA SALINAS	 	
16	ANDRES MARTIN PACHACAMA MARTINEZ		

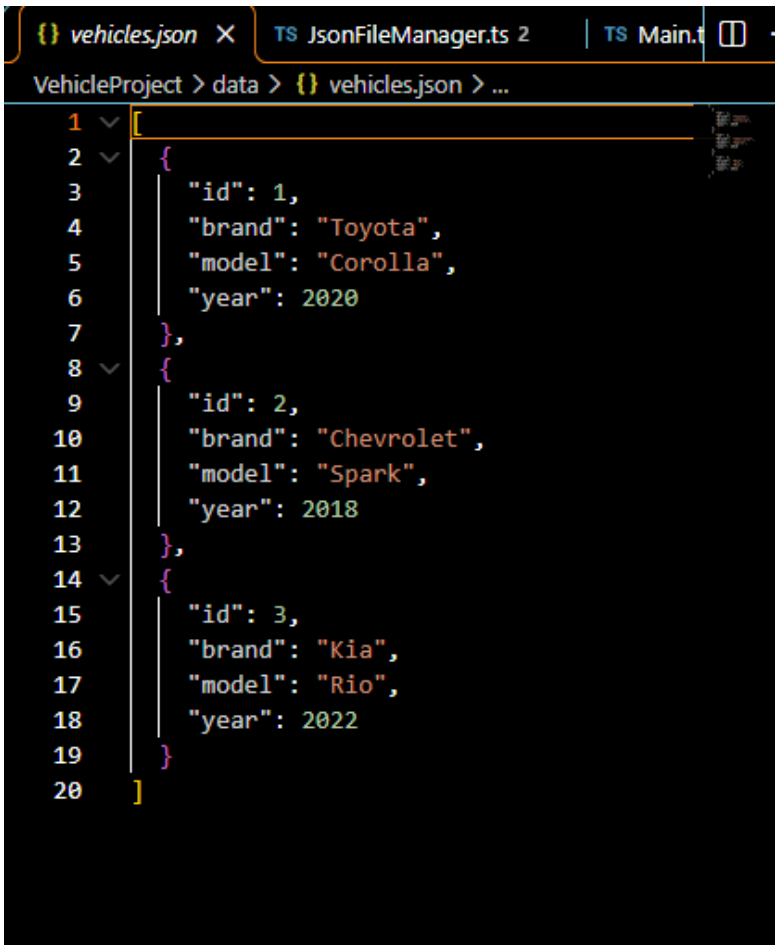

Run Terminal Help
DollApp
class_diagram.mmd
doll.py
json_file_manager.py
{} dolls.json > { } 0 > abc name
[
{
"id": 1,
"name": "Barbie",
"brand": "Mattel",
"price": 29.99
},
{
"id": 2,
"name": "Rapunzel",
"brand": "Disney",
"price": 24.5
},
{
"id": 3,
"name": "Bratz Cloe",
"brand": "MGA",
"price": 19.99
},
{
"id": 4,
"name": "Lucya",
"brand": "Pattys",
"price": 50.3
}
]

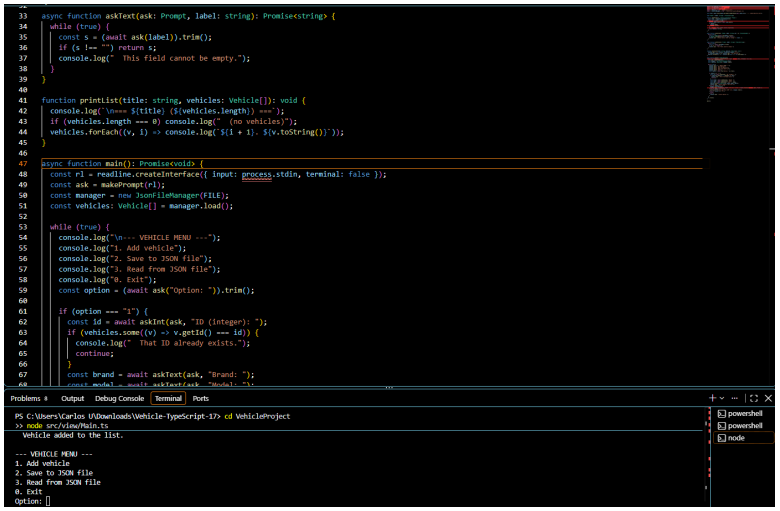

Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\USER\Documents\PythonProjects\DollApp> python main.py
[OK] 3 dolls loaded from dolls.json
ANDRES PACHACAMA'S DOLL APP
ArrayList + JSON file
Dolls in the list: 3 | everything saved
[1] Create dolls
[2] Save dolls to dolls.json
[3] Read dolls from dolls.json
[4] Show dolls in the list
[0] Exit
Choose an option: 1
CREATE DOLLS
Doll #4
id 4
name ..... Lucya
brand ..... Pattys
price ($) 50.3
[OK] Lucya was added to the list (4 dolls)
Add another doll? (y/n): n
Remember to save them with option [2]
Press Enter to go back to the menu...
ANDRES PACHACAMA'S DOLL APP
ArrayList + JSON file
Dolls in the list: 4 | unsaved changes (*)
[1] Create dolls
[2] Save dolls to dolls.json
[3] Read dolls from dolls.json
[4] Show dolls in the list
[0] Exit
Choose an option: 2

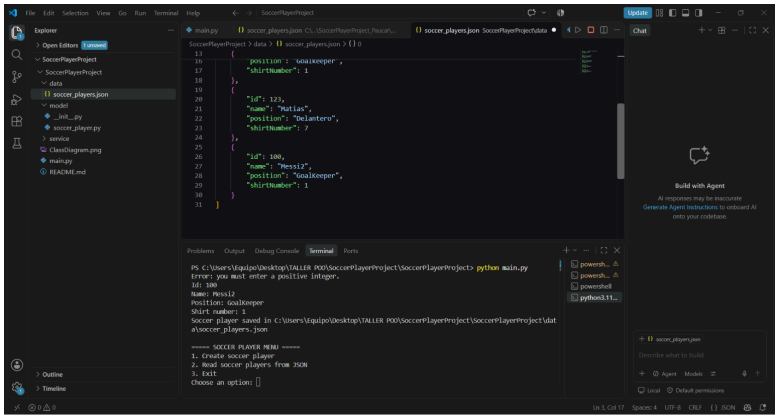
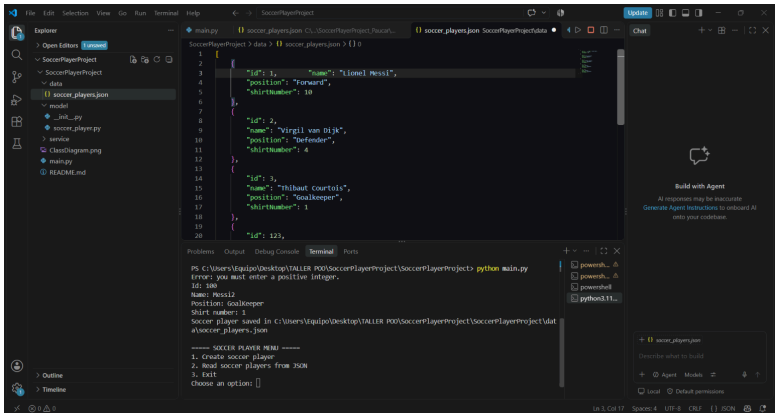
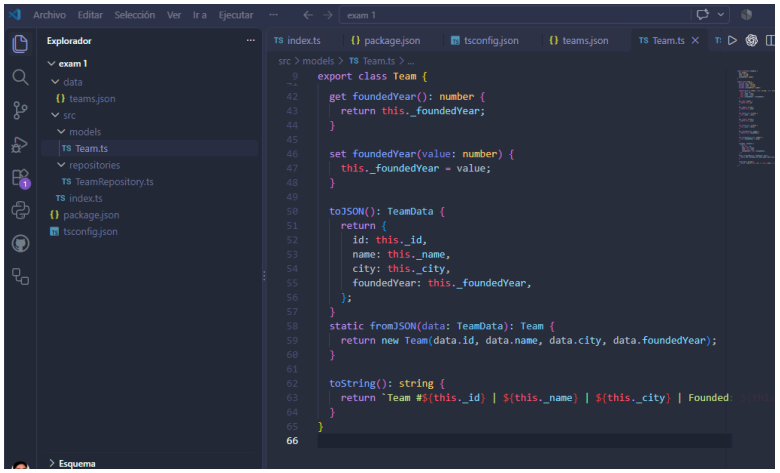

class_diagram.mmd
doll.py
json_file_manager.py
doll_app.py
dolls.json
{} dolls.json > [ ] 0 > abc name
1 [
Problems Output Debug Console Terminal Ports
PS C:\Users\USER\Documents\PythonProjects\DollApp> python main.py
---------- SAVE DOLLS TO dolls.json ----------
[OK] 4 dolls saved in:
C:\Users\USER\Documents\PythonProjects\DollApp\dolls.json
Press Enter to go back to the menu...
ANDRES PACHACAMA'S DOLL APP
ArrayList + JSON file
Dolls in the list: 4 | everything saved
[1] Create dolls
[2] Save dolls to dolls.json
[3] Read dolls from dolls.json
[4] Show dolls in the list
[0] Exit
Choose an option: 3
---------- DOLLS IN dolls.json ----------
ID | NAME | BRAND | PRICE
1 | Barbie | Mattel | $29.99
2 | Rapunzel | Disney | $24.50
3 | Bratz Cloe | MGA | $19.99
4 | Lucya | Pattys | $50.30
Total: 4 dolls
Press Enter to go back to the menu...
ANDRES PACHACAMA'S DOLL APP
ArrayList + JSON file
Dolls in the list: 4 | everything saved
[1] Create dolls
[2] Save dolls to dolls.json
[3] Read dolls from dolls.json
[4] Show dolls in the list
[0] Exit
Choose an option: 4

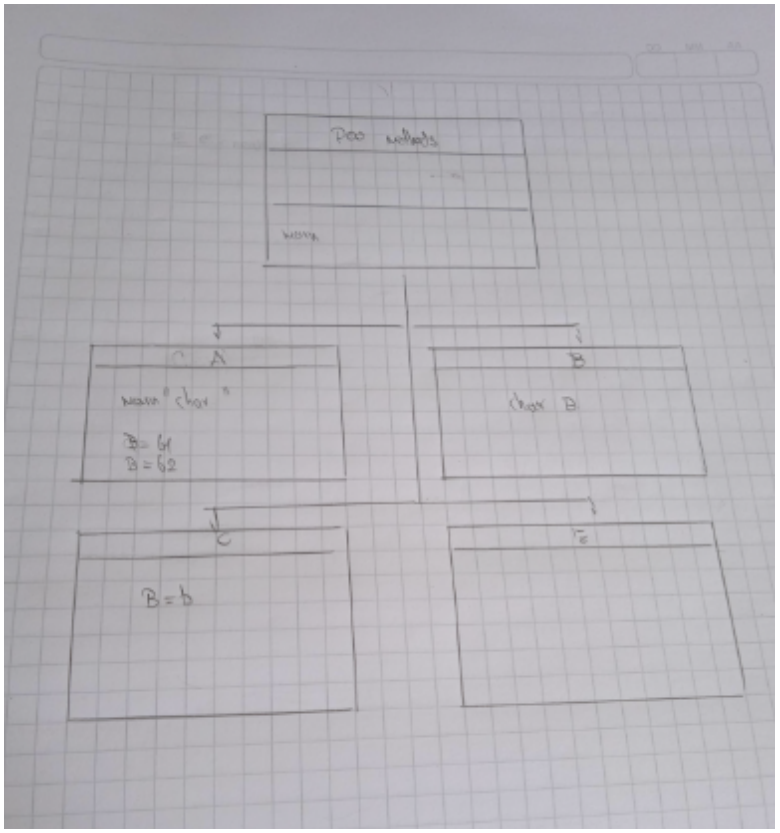
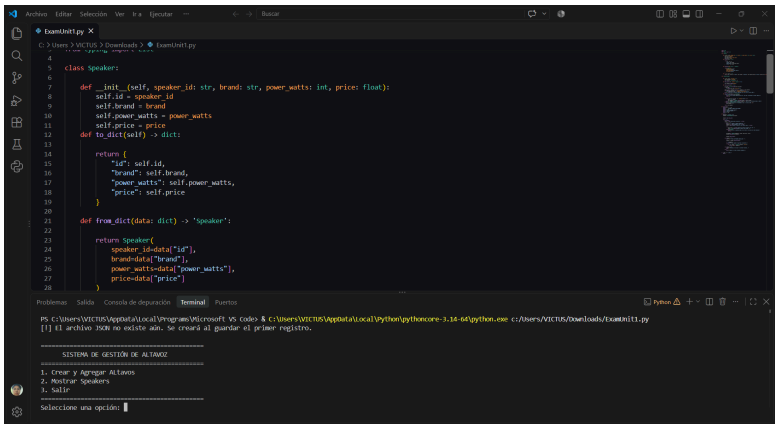
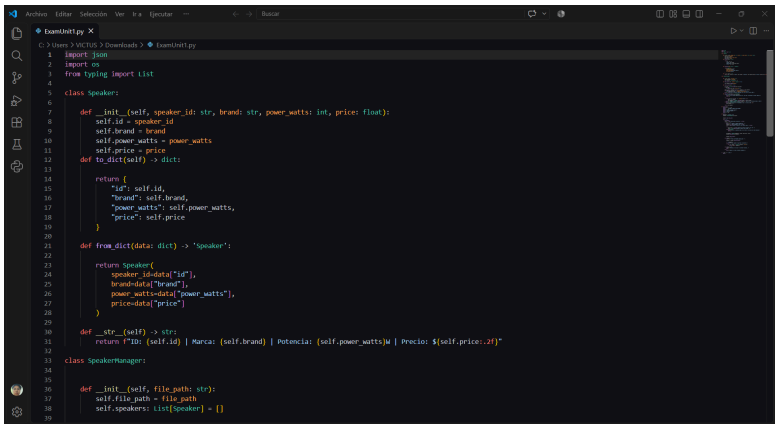

Problems Output Debug Console Terminal Ports
PS C:\Users\USER\Documents\PythonProjects\DollApp> python main.py
---------- DOLLS IN THE LIST ----------
ID | NAME | BRAND | PRICE
1 | Barbie | Mattel | $29.99
2 | Rapunzel | Disney | $24.50
3 | Bratz Cloe | MGA | $19.99
4 | Lucya | Pattys | $50.30
Total: 4 dolls
Doll{id=1, name=Barbie, brand=Mattel, price=29.99}
Doll{id=2, name=Rapunzel, brand=Disney, price=24.5}
Doll{id=3, name=Bratz Cloe, brand=MGA, price=19.99}
Doll{id=4, name=Lucya, brand=Pattys, price=50.3}
Press Enter to go back to the menu...
ANDRES PACHACAMA'S DOLL APP
ArrayList + JSON file
Dolls in the list: 4 | everything saved
[1] Create dolls
[2] Save dolls to dolls.json
[3] Read dolls from dolls.json
[4] Show dolls in the list
[0] Exit
Choose an option: 0
---------- GOODBYE ----------
Thanks for using the Doll App!
PS C:\Users\USER\Documents\PythonProjects\DollApp> python main.py
[OK] 4 dolls loaded from dolls.json

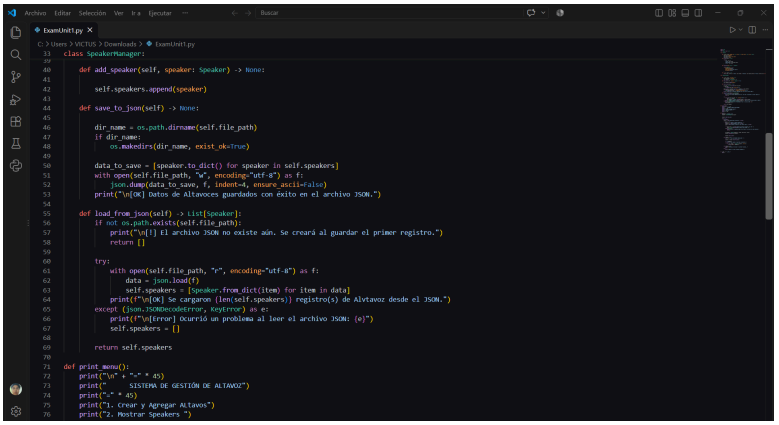
17	CARLOS ALBERTO PARRA AYALA		


vehicles.json
JsonFileManager.ts 2
Main.t
VehicleProject > data > {} vehicles.json > ...
[
{
"id": 1,
"brand": "Toyota",
"model": "Corolla",
"year": 2020
},
{
"id": 2,
"brand": "Chevrolet",
"model": "Spark",
"year": 2018
},
{
"id": 3,
"brand": "Kia",
"model": "Rio",
"year": 2022
}
]


async function askText(ask: Prompt, label: string): Promise<string> {
function printList(title: string, vehicles: Vehicle[]): void {
async function main(): Promise<void> {
Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\Carlos U\Downloads\Vehicle-TypeScript-17> cd VehicleProject
>> node src/view/Main.ts
Vehicle added to the list.
--- VEHICLE MENU ---
1. Add vehicle
2. Save to JSON file
3. Read from JSON file
0. Exit
Option:

18	MATIAS LEANDRO PAUCAR AÑAMISE	 	
19	DENIS ALEXIS PAUCAR COLUMBA	 	

			
20	DANIEL ADRIAN SANGOQUIZA BUENAÑO	 	



```python
class SpeakerManager:

    def add_speaker(self, speaker: Speaker) -> None:

        self.speakers.append(speaker)

    def save_to_json(self) -> None:

        dir_name = os.path.dirname(self.file_path)
        if dir_name:
            os.makedirs(dir_name, exist_ok=True)

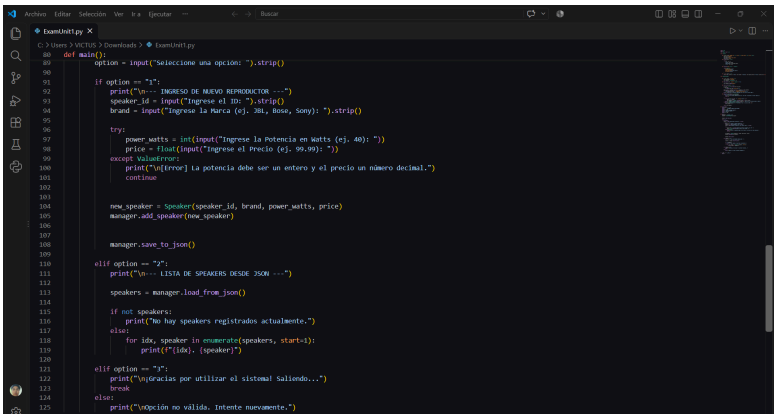
        data_to_save = [speaker.to_dict() for speaker in self.speakers]
        with open(self.file_path, "w", encoding="utf-8") as f:
            json.dump(data_to_save, f, indent=4, ensure_ascii=False)
        print("\n[OK] Datos de Altavoces guardados con exito en el archivo JSON.")

    def load_from_json(self) -> List[Speaker]:
        if not os.path.exists(self.file_path):
            print("\n[!] El archivo JSON no existe aún. Se creará al guardar el primer registro.")
            return []

        try:
            with open(self.file_path, "r", encoding="utf-8") as f:
                data = json.load(f)
                self.speakers = [Speaker.from_dict(item) for item in data]
            print(f"\n[OK] Se cargaron {len(self.speakers)} registro(s) de Alvtavoz desde el JSON.")
        except (json.JSONDecodeError, KeyError) as e:
            print(f"\n[Error] Ocurrió un problema al leer el archivo JSON: {e}")
            self.speakers = []

        return self.speakers

def print_menu():
    print("\n" + "-" * 45)
    print("      SISTEMA DE GESTIÓN DE ALTAVOZ")
    print("-" * 45)
    print("1. Crear y Agregar ALtavos")
    print("2. Mostrar Speakers ")
```



```python
def main():
    option = input("Seleccione una opción: ").strip()

    if option == "1":
        print("\n--- INGRESO DE NUEVO REPRODUCTOR ---")
        speaker_id = input("Ingrese el ID: ").strip()
        brand = input("Ingrese la Marca (ej. JBL, Bose, Sony): ").strip()

        try:
            power_watts = int(input("Ingrese la Potencia en Watts (ej. 40): "))
            price = float(input("Ingrese el Precio (ej. 99.99): "))
        except ValueError:
            print("\n[Error] La potencia debe ser un entero y el precio un número decimal.")
            continue


        new_speaker = Speaker(speaker_id, brand, power_watts, price)
        manager.add_speaker(new_speaker)

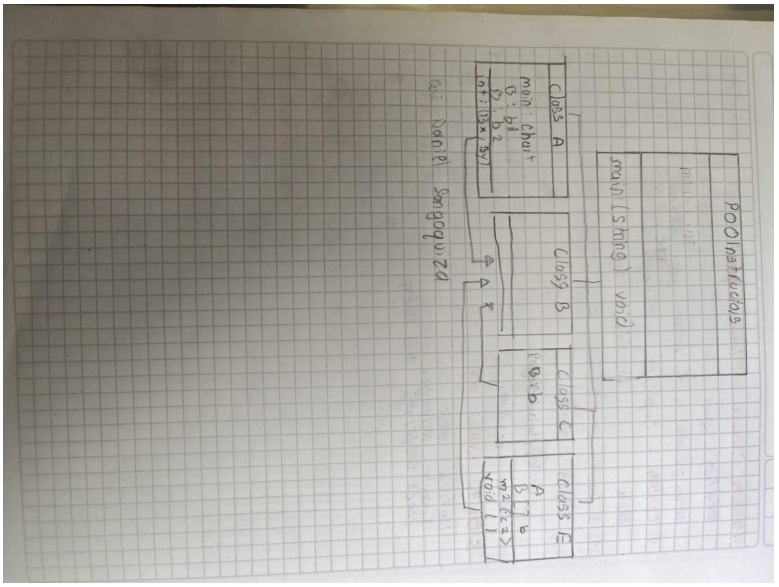

        manager.save_to_json()

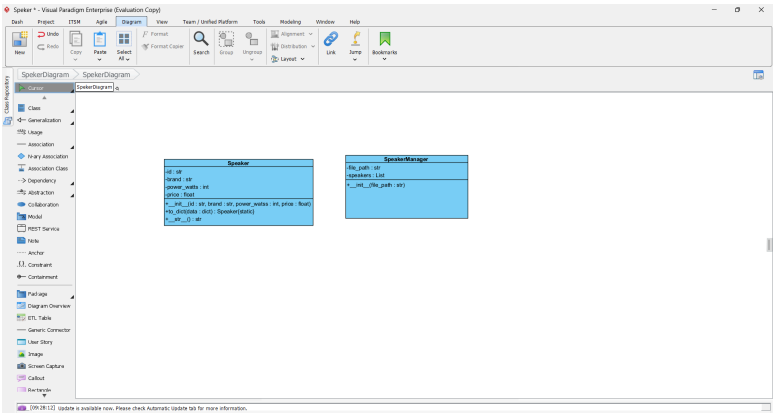
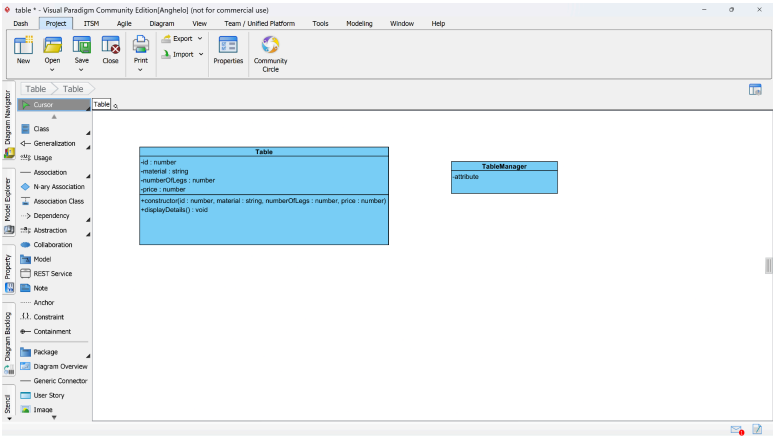
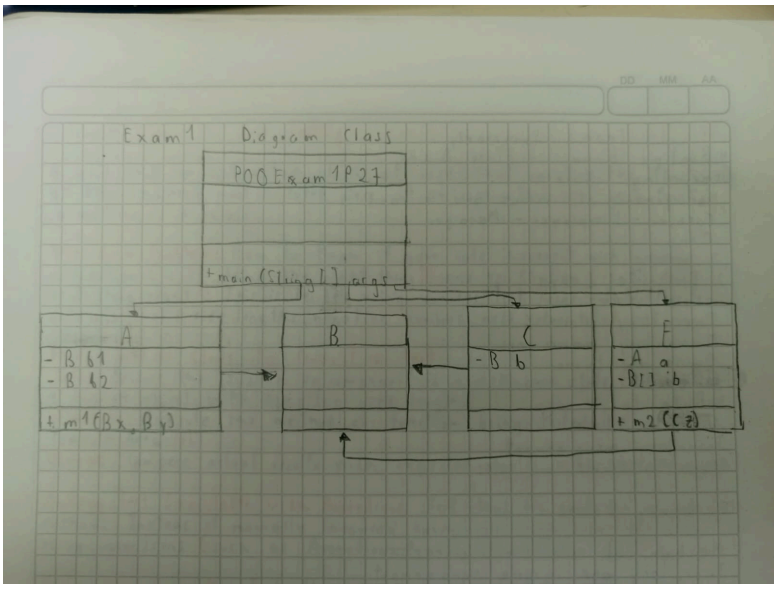
    elif option == "2":
        print("\n--- LISTA DE SPEAKERS DESDE JSON ---")

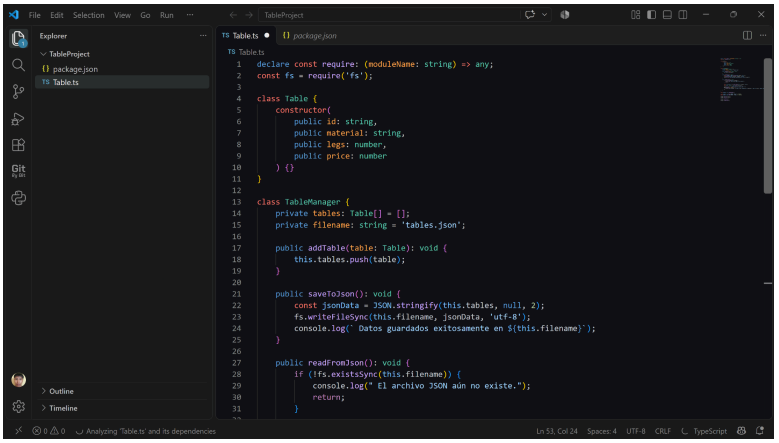
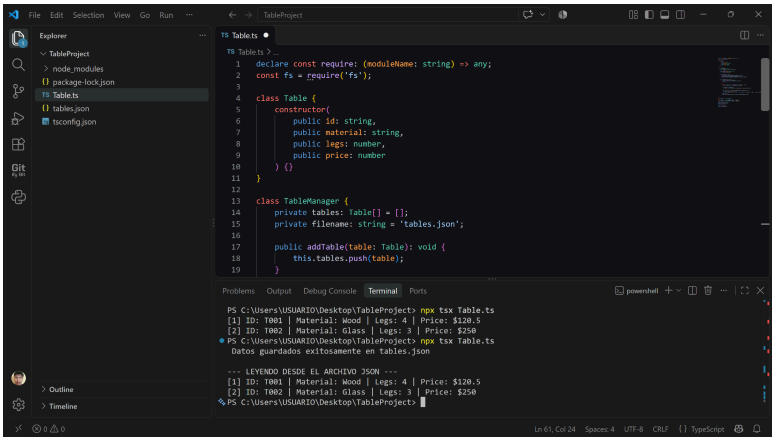
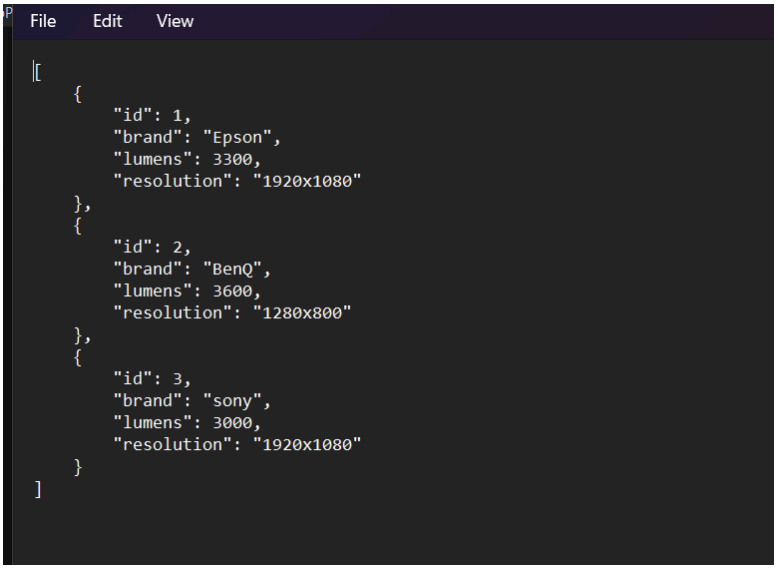
        speakers = manager.load_from_json()

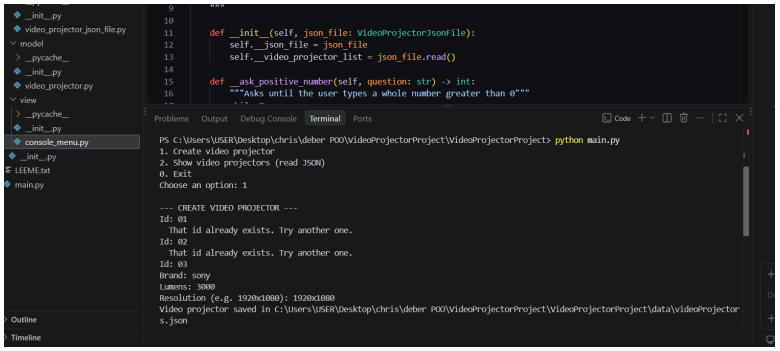
        if not speakers:
            print("No hay speakers registrados actualmente.")
        else:
            for idx, speaker in enumerate(speakers, start=1):
                print(f"{idx}. {speaker}")

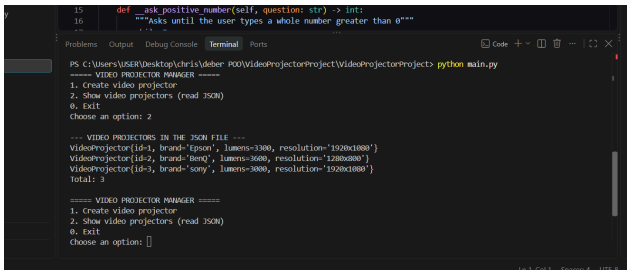
    elif option == "3":
        print("\n¡Gracias por utilizar el sistema! Saliendo...")
        break
    else:
        print("\nOpción no válida. Intente nuevamente.")
```

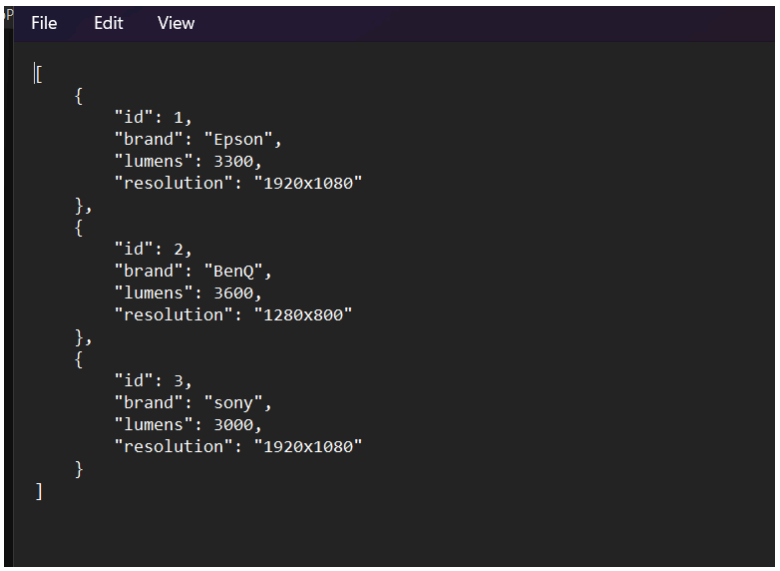


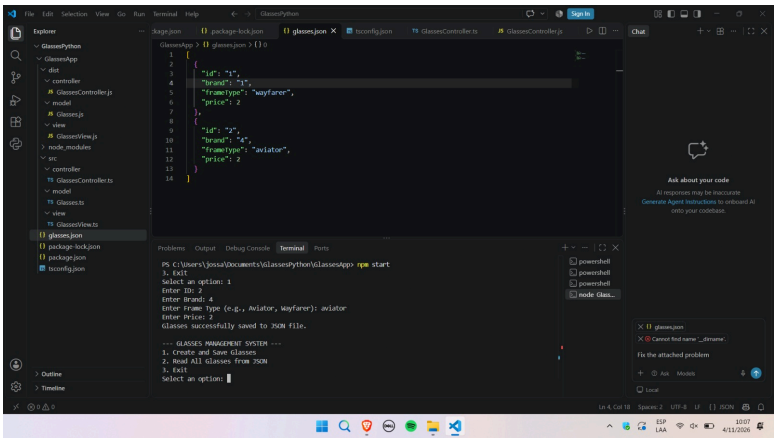
			
21	ANGHELO JAVIER SUNTAXI CRUZ	 	

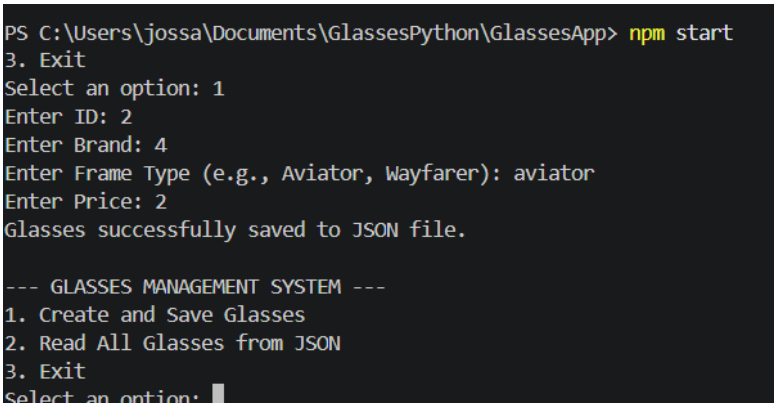
		 	
22	OSTIN CHRISTOPHER TORRES GALEAS		


__init__.py
video_projector_json_file.py
model
__pycache__
__init__.py
video_projector.py
view
__pycache__
__init__.py
console_menu.py
__init__.py
LEEME.txt
main.py
Outline
Timeline
"""
def __init__(self, json_file: VideoProjectorJsonFile):
self.__json_file = json_file
self.__video_projector_list = json_file.read()
def __ask_positive_number(self, question: str) -> int:
"""Asks until the user types a whole number greater than 0"""
Problems Output Debug Console Terminal Ports
Code
PS C:\Users\USER\Desktop\chris\deber POO\VideoProjectorProject\VideoProjectorProject> python main.py
1. Create video projector
2. Show video projectors (read JSON)
0. Exit
Choose an option: 1
--- CREATE VIDEO PROJECTOR ---
Id: 01
That id already exists. Try another one.
Id: 02
That id already exists. Try another one.
Id: 03
Brand: sony
Lumens: 3000
Resolution (e.g. 1920x1080): 1920x1080
Video projector saved in C:\Users\USER\Desktop\chris\deber POO\VideoProjectorProject\VideoProjectorProject\data\videoProjector
s.json

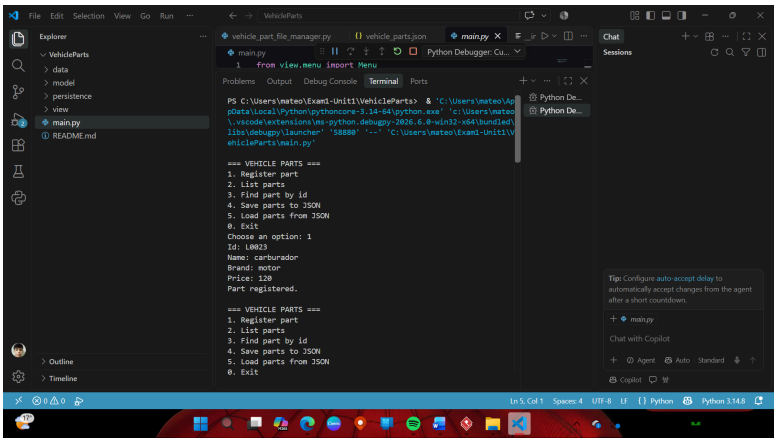
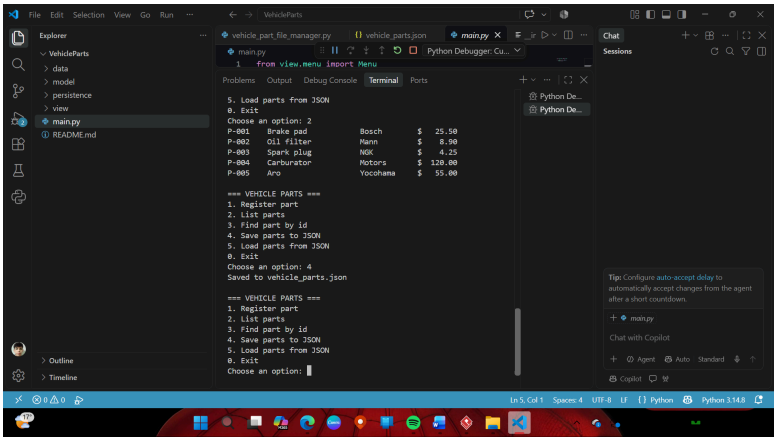
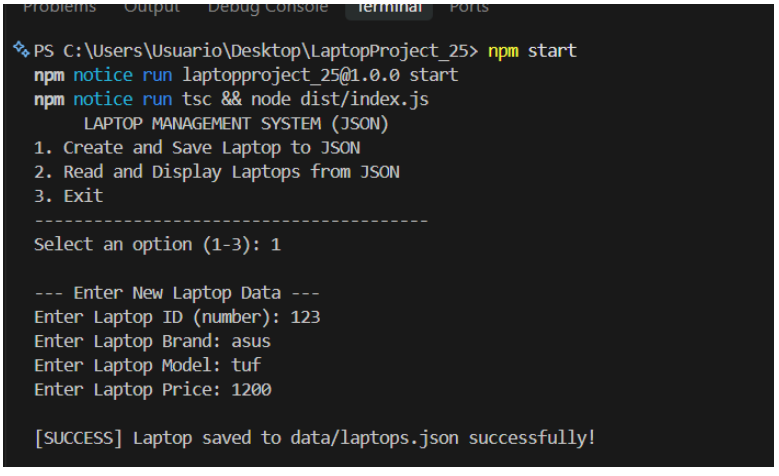

def __ask_positive_number(self, question: str) -> int:
"""Asks until the user types a whole number greater than 0"""
Problems Output Debug Console Terminal Ports
Code
PS C:\Users\USER\Desktop\chris\deber POO\VideoProjectorProject\VideoProjectorProject> python main.py
===== VIDEO PROJECTOR MANAGER =====
1. Create video projector
2. Show video projectors (read JSON)
0. Exit
Choose an option: 2
--- VIDEO PROJECTORS IN THE JSON FILE ---
VideoProjector{id=1, brand='Epson', lumens=3300, resolution='1920x1080'}
VideoProjector{id=2, brand='BenQ', lumens=3600, resolution='1280x800'}
VideoProjector{id=3, brand='sony', lumens=3000, resolution='1920x1080'}
Total: 3
===== VIDEO PROJECTOR MANAGER =====
1. Create video projector
2. Show video projectors (read JSON)
0. Exit
Choose an option:


File Edit View
[
{
"id": 1,
"brand": "Epson",
"lumens": 3300,
"resolution": "1920x1080"
},
{
"id": 2,
"brand": "BenQ",
"lumens": 3600,
"resolution": "1280x800"
},
{
"id": 3,
"brand": "sony",
"lumens": 3000,
"resolution": "1920x1080"
}
]

23	JOSSELYN ANDREA ULCUANGO QUIMBIAMBA		



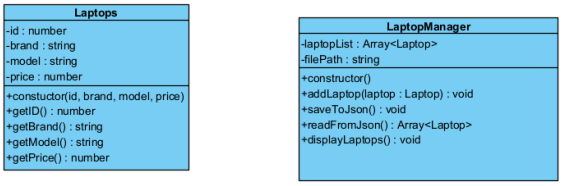


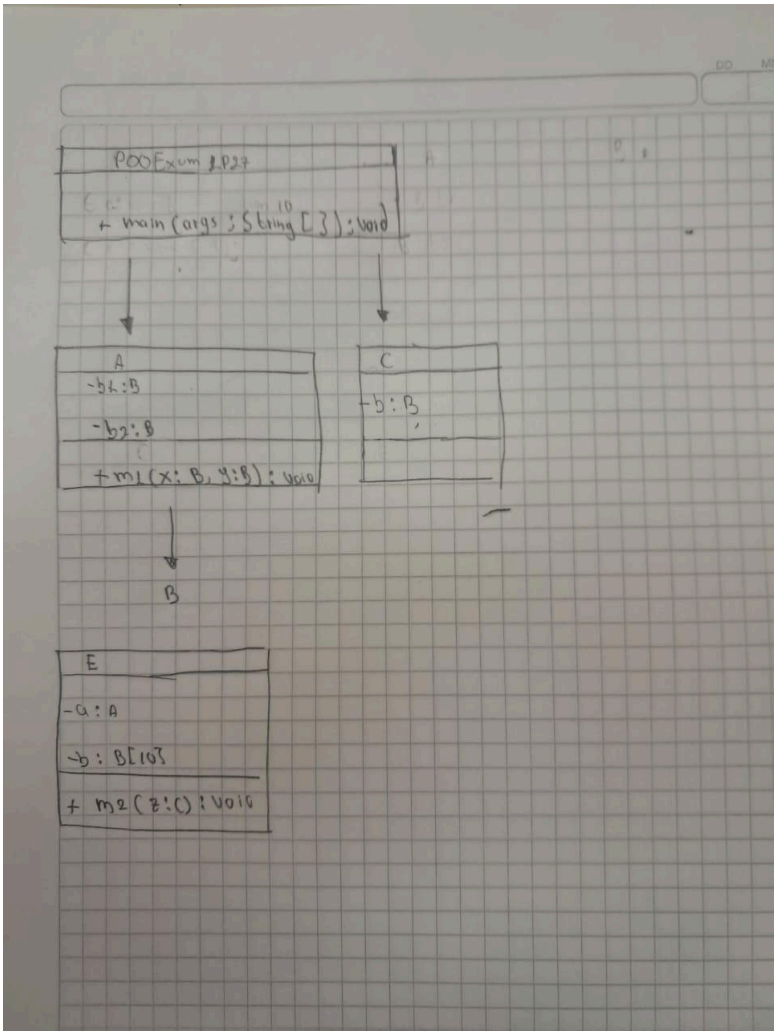
24	MATEO JHAER UNDA PIGUAVE	  	
25	YANDRY JOSUE VELEZ VALENCIA	 	

```
     LAPTOP MANAGEMENT SYSTEM (JSON)
1. Create and Save Laptop to JSON
2. Read and Display Laptops from JSON
3. Exit
---------------------------------------
Select an option (1-3): 2

========== LAPTOPS IN JSON FILE (ARRAYLIST) ==========
1. ID: 123 | Brand: asus | Model: tuf | Price: $1200.0
=======================================================

     LAPTOP MANAGEMENT SYSTEM (JSON)
1. Create and Save Laptop to JSON
2. Read and Display Laptops from JSON
3. Exit
---------------------------------------
Select an option (1-3):
```




POOExam1P27
+ main (args : String[]) : void
A
-b1 : B
-b2 : B
+ m1 (x : B, y : B) : void
B
C
-b : B
E
-a : A
-b : B[10]
+ m2 (z : C) : void